TFPT — The Prime Front

Research documentation of the prime / zeta line: seven verified modules, one localized residual, and the complete kill list

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July 28, 2026 · v5.4 (rev 425)

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*

1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.

2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.

3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.

4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\mathrm{geo}}, G_{\mathrm{net}}, F_{\mathrm{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** `tfpt_prime_front` — research documentation of the prime / zeta line: the seven load-bearing modules on the E_8 glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

*You are reading the **Prime Front** companion (P): the complete internal record of the prime / zeta line — how a discrete compiler's bookkeeping kept producing classical number-theoretic objects, which of those links became load-bearing modules, and how the remaining question was compressed, over a hundred exploratory probes, down to one strict-positivity margin input — then, once that margin was measured to be an artifact of the requirement, down to a two-object core — then down to a three-point list of which only one point is an inequality — and finally, once the boundary reformulation ran, down to that single inequality in a textbook shape (a Galerkin error with Aubin–Nitsche duality), with the prime comb named as the one structure that refuses compression — and, once the missing lemmas of that inequality were measured against their classical addresses, down to exactly one genuinely new analytic statement. It is a **research documentation**, not a results paper. Its purpose is that a reader (or a reviewer) can see the whole route, including everything that died on the way.*

Abstract

A two-axiom discrete compiler ($c_3 = 1/(8\pi)$, $g_{\mathrm{car}} = 5$, $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$) has no business meeting the prime numbers. It does anyway: the μ_4 glue completion carries a quartic character,

the signed glue census is a theta series, and the shell counts of that census turn out to be *Hecke* data. It goes further than an analogy: the E_8 shell count is literally $240\,\zeta(s)\zeta(s-3)$ as a Dirichlet series, and the lattice's self-duality gives its theta sum an exact functional equation whose fixed axis is the critical line — a framework picture, stated as such, which the document opens with and then keeps at arm's length. This companion documents the line end to end. Seven modules are load-bearing and ledger-carried (v535–v541): Hecke from geometry on the E_8 census, the Eichler trace layer, the half-integral bridge, one finite relative-trace identity of which the first three are projections, the Weil structure of the resulting family with two explicitly isolated obstructions, the amplitude route with a positive linear carrier, and the matching-lemma / transport ledger package. Everything beyond those seven is *sandbox*: an exploratory program, run as 118 preregistered probes with 2968 machine checks, which localized the residual object. That localization is the actual content of this document. The transport ledger closes exactly, leaving one coupled inequality (called **I5**) which is — *by identity, as a typing and not as progress* — equivalent to Weil positivity and hence to the Riemann Hypothesis. The subsequent arc maps I5 geographically (Connes–Consani dictionary, tightness curves, the Sonin crossing, the one-atom band), turns the positivity question into a per-zone handover induction, and then certifies its ingredients one by one until the entire remaining hardness sits in a single Loewner statement on the J-odd half of a window, reducible by Sherman–Morrison/Woodbury to one scalar ratio $r = \kappa/\varepsilon \le 1$ (measured 0.0053–0.1814 on 16 zones), then, once ε is identified as an exact energy, to two scalars — and finally, with both scalars certified (the wing scalar unconditionally by a graded matrix cap, the boundary value by a residual certificate that carries a cancellation), to a chain that closes 16/16 conditional on exactly one strict-positivity margin input, itself two to six orders of magnitude weaker than the conclusion it buys. The penultimate part then closes the induction circle on the measured zones end to end — a certified base case, fifteen certified handover steps through a graded Loewner minorant, and the structural finding that the atom entry costs nothing — while leaving three sharp, named gaps: no reserve, no scalar step law, and no uniformity in k, with an extrapolated crossing warning at $n \approx 170$. The next part stops extrapolating and measures: on a deep ladder of 199 prime-power zones with 117 certified handovers, the crossing is located between $n = 461$ and $n = 463$ — the $n \approx 170$ figure was a fit artefact, but the wall is real, and $n^* \approx 462$ is an upper bound — and the single warning splits into three separate walls (a measured margin wall, an arithmetic ladder wall set by the twin-prime pair $521 \to 523$, and a vacuous-requirement wall at $n = 727$), while the handover mechanism itself never fails: all 117 steps certify at retention 1.000000, and the chain tears at the ratio, not at a step. The operating variable turns out to be the window depth, not the zone index. The next part acts on that reinterpretation and rebuilds the construction in a gap-coupled scaled frame (D tied to the local prime gap): two of the three walls fall *structurally* — the ladder death becomes a payable cell cost (the killer pair $461 \to 463$ now certifies spectrally with the reserve opening by a factor $\sim$1000) and the requirement wall loses its depth dependence — while the margin wall proves *frame-invariant* at exponent -0.974: it is the substance of the requirement, not geometry. The hardness is thereby traded — three arithmetic walls against two analytic demands, the positivity of one limit operator and a convergence rate below the step reserve — with the prime-gap dependence disclosed rather than hidden. The next part settles the currency question, with a fundamental reinterpretation: the wall carries the same exponent in *every* admissible currency — it is substance — but it measures the *discretisation*, not the spectrum. Under refinement at a fixed window both leading eigenvalues carry the same cell-width power (the continuum window form has no gap), the positive floor survives only as a cancellation of relative size 10^{-7}–10^{-4}, and the T109 requirement chain turns out to divide by an artifact margin; the hardness balance is restated as four parts, led by a margin-free step certificate. The next part builds that certificate and the wall dissolves: the exact Schur complement (Albert 1969, with the Moore–Penrose inverse) certifies every ladder step margin-free — eleven of them beyond the old wall, to $n = 1331$, and all seven zones where the requirement chain tore — the wall was an $O(1)$ numerator divided by an artifact floor, chains now stop only at the compute cap and never at a step, and the (R) demand itself proves grid-bound and is deleted rather than transported, leaving a two-object core: ε relative to κ via the exact Szegő identity, and transport of an $O(0.1)$ Schur complement between grids. The next part splits that core: the transport across a real regrid certifies only mild refinement (a clean split at ratio $\rho^* \approx 1.8$, and for a principled reason — on nested ladders, where the transport error is exactly zero, the Schur floor itself falls like $\rho^{-1.7}$, so no bound can undo the drop), while a two-scale compression breaks the compute cap: a certified margin-free step at $n = 155\,921$ (117× deeper), chains of ten steps, the stopper always cost and never a step — leaving the shortest list the program has produced: three points, of which only one is an inequality (the classical Szegő–Levinson prediction-error bound for one symbol). The next part executes the reformulation point of that list, and the induction step turns out to *be* a boundary process: partial minimisation over the interior (Haynsworth) compresses the window into a state (Σ, r, σ) of $12 \times 12 + 12 + 1$ numbers that carries the global rank-one pole *exactly* (bordered elimination, no truncation) and updates by a Riccati-type recursion whose cost is independent of the window — one unbroken chain of 169 236 prepends ran to a window of 1.35 million cells (903× the old cap) at a flat 76 μs per step, declared honestly as a cost-geometry demonstration and not a Weil certificate. What refuses compression is the *prime comb* itself: the archimedean tail decays cleanly, but the reflection comb couples every incoming cell to an interior cell at $\alpha - \log n_j$ for every prime power in the window, so the full symbol never decays in the width and no sparse faithful state exists. In exchange, the one remaining inequality acquires a textbook shape: ε is the error of a piecewise-constant Galerkin method for the D-independent function $2\sinh(x/2)$, Aubin–Nitsche duality predicts the measured exponent ($\theta = 1.79 \Rightarrow s \approx 0.90$), and the target inequality is $\varepsilon(\alpha, D) \geq c_0 D^{\theta} \alpha^{\varphi}$ across the prime-power jumps. The last part then makes that inequality *theorem-shaped*: the Galerkin reading becomes four exact identities, the lower-bound direction acquires a certified two-level chain that loses no power of D ($\theta' = 1.74$ against $\theta = 1.76$), the jumps acquire exact closed forms — with the correction that prime-power entries *raise* ε, so the factor-120 falls were a sweep artefact of the earlier run — and what remains is three named analytic lemmas about one symbol, two of them constants rather than rates. The closing part measures those three lemmas against their classical addresses, and two of the three stand: saturation is an *identity* here; the Schur-floor lemma is rescued by the strengthened Cauchy–Schwarz shift onto the oscillation Gram, whose arithmetic aliasing symbol suppresses the prime comb quadratically — certified positive on half the measured windows, and on a 15 680-point FFT lever the certified floor rises logarithmically and crosses zero, so the failing windows were under-resolved rather than obstructed; and the corner lemma corrects itself — a log singularity is the boundary of all powers, so no power-law corner exponent exists to cite, and the certified chain is repaired at its last step, the rate now living on the oscillation mass. The missing analytic content is thereby exactly one statement: a $D^{1.75}$ lower bound for that mass. **No claim of progress on the Riemann Hypothesis is made anywhere in this document**, and none of the sandbox findings moves any ledger marker. The document is deliberately symmetric about failure: a dedicated section lists the readings and routes that were refuted — an essential-singularity reading, a C/g proxy law, a density chain, an accumulated invariant with self-propagation, the Landau–Widom anchor, the symbol route for the odd channel, three certificate candidates for the avoidance norm plus the Combes–Thomas certificate at the boundary, and the naive margin route — because in a program of this shape the kill list is the more reliable output.

[E] exact / machine-proven **[C]** conditional **[O]** open / axiom **[X]** kill test

1 How to read this document

Scope and status — read this before anything else

1. **No RH claim.** Nothing in this document claims progress towards, or evidence for, the Riemann Hypothesis. Where an equivalence to RH appears (Section 5) it is an *equivalence typing*: a statement about which object carries the difficulty, produced by an exact identity, and deliberately *not* a statement that the difficulty has been reduced.
2. **Two layers, never mixed.** Claims marked [ledger] are load-bearing: they have a row in `verification/status_ledger.csv`, a module in `verification/`, and a script citation here (the blue `[verification/...]` tag). Everything marked [sandbox] is exploratory work in `experiments/tfpt-discovery/`; it carries no ledger row, no marker, and no authority. Section 4 is the only section that is entirely [ledger].
3. **Status markers are quoted, never upgraded.** The markers **[E]** **[C]** **[O]** **[X]** shown in Section 4 are read verbatim from the ledger rows of v535–v541. This document introduces no new markers and moves none.
4. **Fits are declared as fits.** Every scaling law, slope, exponent and extrapolation quoted from the sandbox is a fit to finitely many measured windows and is labelled as such at the point of use. Sixteen resolved zones are sixteen zones, not all zones.
5. **Classical results are named as classical.** Kneser neighbours, Eichler/Witt, Shimura, Waldspurger, Cohen, Weil/Guinand, Bombieri, Yoshida, Connes–Consani, Gronwall/Robin, Douglas, Slepian–Landau–Pollak, Sherman–Morrison/Woodbury, Grenander–Szegő and the rest are classical and are used as such; the compiler-native part is the wiring, not the theorems.
6. **Numbers are literal.** Every numeric value in this document is transcribed from the research diary `experiments/next.txt` or from a ledger row. Figures that could not be drawn from literal data are marked *schematic* in their caption, and the caption states exactly which quantities in the figure are literal.

The diary parts are cited as **T**n (part n of the research diary, chronological, $n = 11 \dots 118$). The probe file, the contract name, the verdict and the check count of every part are listed in Appendix A. Section 2 explains why a physics compiler meets the primes at all and Section 3 states the framework interpretation that motivated the whole line — a self-dual lattice, its functional equation, and a mirror that returns at the far end of the program; Section 4 is the verified layer; Sections 5, 6 and 7 are the sandbox program in three acts; Section 8 is the kill list; Section 9 is the status statement.

2 Motivation: why a compiler meets the primes

TFPT is a discrete compiler with two inputs: the seam constant $c_3 = \frac{1}{8\pi}$ (P1) and the carrier rank $g_{\mathrm{car}} = 5$ (P2). They force the lattice completion $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, and the readouts of the Standard-Model skeleton are taken off that completion. None of this is about number theory. The prime line begins with an accident of the *glue.*

The completion is not $D_5 \oplus A_3$ but $D_5 \oplus A_3$ *plus* a glue group μ_4 — a cyclic group of order four. A cyclic group of order four carries a quartic character, and the quartic character on $\mathbb{Z}$ is χ_{-4}, the non-trivial character mod 4. Once the census of glue vectors is counted *with that sign*, three things happen at once, and all three are elementary consequences rather than choices:

1. The signed glue count is a **theta series**. Signed counting of a quadratic form is a theta series by definition; here the compiler's own count θ_3^2 is exactly the Gaussian-integer theta, $r_2(n)$, so the census counts $\mathbb{Z}[i]$-ideals (T85).
2. The character makes the count a **Dirichlet object**. χ_{-4} is the character of $\mathbb{Q}(i)$; the associated L-function is the first Dirichlet L-function, and the split law "$p = a^2 + b^2$" becomes a compiler-visible fibre property of each prime (T21, channel B: χ_4 fibre $\iff p = a^2 + b^2$, verified 81/0/87/0 on $p < 1000$; χ_3 fibre $\iff p = a^2 + 3b^2$, 80/0/86/0).

3. The shell counts of the census carry **Hecke eigenvalues**. That is the surprise, and it is the content of the first load-bearing module.

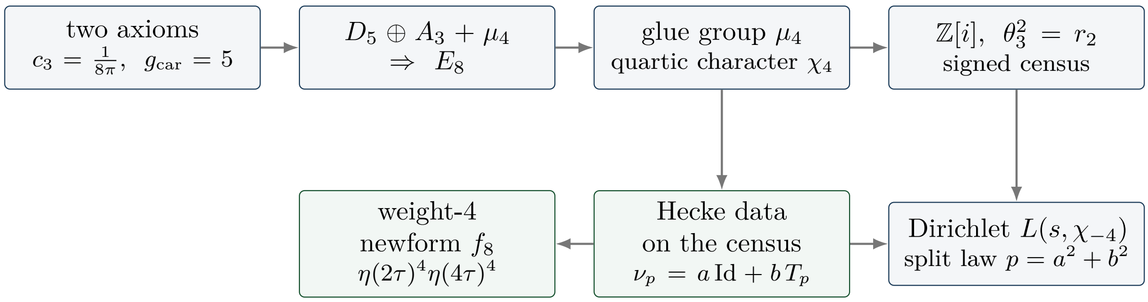

Figure 1: The chain by which the compiler acquires arithmetic. Nothing here is chosen for number theory: the glue group is forced by the completion, the quartic character is the character of a cyclic group of order four, and the signed count of a quadratic form is a theta series. The green nodes are where the line becomes load-bearing (Section 4).

The first question of the series was therefore not "does this prove anything about primes" but "which of these links are *mechanism* and which are pretty coincidences". Parts T11–T13 established the signed glue theta, its L-series and the glue-character atlas (19/19, 16/16, 11/11); T14–T19 tested six candidate contracts in parallel and produced two self-corrections against the project's own earlier readings (T14 DEFLATED: the measured 2π is the universal Bisognano–Wichmann/Unruh conversion, not "the self-dual width"). The cross-finding of that batch is worth stating because it survived the entire series: the structure is exact at the census/lattice level (weight 4), and the gap is always the *same* gap — the canonical transport to the GL(1) / $\xi(s)$ world (multiplicity one, weight 1/2, the archimedean place, the derivation of the Hecke action). One meta-contract, `ZETA.CENSUS.TO.GL1` [O].

Parts T20–T25 then killed, in four stages, the slogan "the archimedean term comes from the seam": the raw seam density of states is flat/slightly decreasing (loglog slope ≈ -0.40) while the classical digamma kernel grows logarithmically (measured 0.15917 against $1/(2\pi) = 0.159155$, rms 3×10^{-6} for $t \geq 20$) — KILLED-AS-NAIVE (T20, 25/25); the non-DOS successor located the missing logarithm in the interval cut but failed the one-constant dictionary at $2/\pi$ (T22, PARTIAL); and both dictionary closures plus the scattering-phase observable died (T23–T25, $23/23 + 25/25 + 29/29$). This matters for the rest of the document: the archimedean place was never going to come for free, and when it finally arrived (T82) it arrived from *inside* the same family, not from the seam.

3 The big picture: a self-dual lattice and its own mirror

What this section is

A **framework interpretation**, not a result. It says how the pieces fit together and why the line was pursued for a hundred parts. Every factual statement it uses is either classical (and named as such), one of the seven modules of Section 4, or a sandbox probe with its part number. Nothing here is an argument about the Riemann Hypothesis, nothing here is used later as a premise, and the picture itself carries no status marker.

3.1 (i) The zeta function is literally in the geometry

Count the E_8 lattice points shell by shell. The even shells satisfy $N(2n) = 240\,\sigma_3(n)$ — direct lattice counting gives 240, 2160, 6720, 17 520, 30 240, 60 480 for $n = 1 \ldots 6$ (T6, one of the parts that predate the indexed range of Appendix A) — and σ_3 is multiplicative, so the counting function of the lattice is a Dirichlet series with an Euler product:

$$\sum_{n\geq 1}\frac{N(2n)}{n^s} \;=\; 240\sum_{n\geq 1}\frac{\sigma_3(n)}{n^s} \;=\; 240\,\zeta(s)\,\zeta(s-3) \qquad (\text{at } s=5:\ 1.705672 \text{ vs } 1.705678).$$

The Riemann zeta function stands in E_8's own counting function, with a simple pole at $s=4$ of residue $240\,\zeta(4)=8\pi^4/3$ (T12). This is classical lattice theory — Eisenstein series of weight 4 — and it is stated here only to fix where the arithmetic enters: not through a modelling choice, but through the object the compiler already had. The contrast is instructive: the same multiplicativity *fails* for the $D_5\oplus A_3$ sublattice alone (52, 576, 1584, 16 128 with $576\cdot 1584\neq 16\,128\cdot 52$, T6). Without the glue there is no Euler structure in the count.

3.2 (ii) The glue carries the character of the Gaussian integers

The completion is $D_5\oplus A_3+\mu_4$, and a cyclic group of order four carries a quartic character, which on $\mathbb{Z}$ is χ_{-4} — the character of $\mathbb{Q}(i)$. Two consequences make the primes a *fibre property* of the compiler rather than an application of it. First, the split $p\equiv 1$ versus $p\equiv 3 \pmod 4$ is visible to the census directly: the χ_4 fibre holds exactly for $p=a^2+b^2$ (81/0/87/0 on $p<1000$, T21). Second, the character kills a pole. Of the three character channels of the E_8 counting function, the trivial one is $240\zeta(s)\zeta(s-3)$ with its pole at $s=4$, the spinor one is $64(1-2^{-s})\zeta\zeta$ and keeps it, while the μ_4-signed channel $16\cdot 2^{-s}\eta(s)\eta(s-3)-8L(f_8,s)$ is *entire* ($L_c(4.001)=-6.976$, finite; T12). The signed channel is the only pole-free one — exactly the ζ versus $L(\chi)$ pattern.

3.3 (iii) Hecke structure is an output of geometry, not an input

The step that made the line load-bearing is `v535`: the Hecke action on the census is produced by *pure lattice adjacency* — Kneser p-neighbours — with no modular input anywhere in the derivation. The marked neighbour sum is exactly an affine Hecke element, $\nu_p=a\,\mathrm{Id}+b\,T_p$, and the cusp eigenvalues come out of the glue census as $a_p=-c(p)/8$. Prime structure is what the geometry *emits*; nothing about primes was put in. The first four modules of Section 4 make this precise and end in one finite relative-trace identity of which the first three are projections.

3.4 (iv) The core point: self-duality, and a mirror that keeps coming back

E_8 is unimodular — self-dual — and self-duality is exactly the hypothesis of Poisson summation. The lattice theta sum therefore satisfies an exact functional equation, $\sum_x e^{-t|x|^2}=(\pi/t)^4\sum_x e^{-(\pi^2/t)|x|^2}$ (verified to 25 digits, T12), whose unique self-dual width is $t=\pi$. A functional equation is a *reflection*, and the critical line is the fixed axis of that reflection. That is the geometric content of the picture: on the axis sits the symmetry, and beside it sits everything the lattice actually carries.

Two findings, at opposite ends of the program, put the same mirror on the table twice.

- **The reflection product is the transport operator** (T83, `fe_symmetrization_probe.py`, 27/27, INVARIANT-NULL). The involution algebra was computed exactly: ζ's functional equation is the reflection $J_{1/2}$ and is absorbed in evenness (on autocorrelations it acts as the identity), while the family's functional equation is J_1. Their *product* $J_1\circ J_{1/2}=e^{\pm u}$ is the unit-line shift — the difference of the two centres *is* the transport operator, algebraically. The two reflections generate an infinite dihedral ladder; the value side is invariant under the whole ladder, the spectral cone only under its own reflection, which is what anchors it to the critical line; and $\mathrm{Fix}(J_1)\cap\{\text{Weil cone}\}=\{0\}$ exactly, so no symmetrisation can relieve I5. The verdict was a null with a structural finding: the wall is *transversal* to the functional equation, not positional.
- **The same mirror sits in the last hard residue** (T106, Section 7). The mirror parity J with $JQJ=Q$ and $JB=-BR$ splits the Weil pole exactly into $ss^{\mathsf{T}}-tt^{\mathsf{T}}$: a positive rank-one lift in the J-even channel and a negative rank-one pressure in the J-odd one. The even channel closes; everything that is still hard lives in the odd half — the anti-symmetric part of the same reflection.

Stated as a picture and nothing more: the recurring question of this line is whether the self-dual object is entirely at peace with its own mirror. That sentence is an interpretation of where the difficulty sits, in the same spirit as the classical reading of the critical line as a symmetry axis. It is *not* a reformulation of the Riemann Hypothesis, it is not used as a premise anywhere below, and it produces no claim.

3.5 (v) What the program did with the picture

It converted it, step by step, into smaller and more explicit objects. The transport ledger closes exactly (v541), which leaves the single coupled inequality I5; I5 is by identity equivalent to Weil positivity, an equivalence *typing* and not a reduction (Section 5); the positivity question is then reorganised as a per-zone handover induction (Section 6); the induction is certified ingredient by ingredient until only one Loewner statement (R) on the J-odd half of a window remains; and (R) reduces first to one scalar ratio, then to two scalars, and finally — with both scalars certified — to one strict-positivity margin input (Section 7). The full cascade, with the type of every arrow marked, is Figure 8; the reason it is a route and not a proof sketch is stated there and in Section 9.

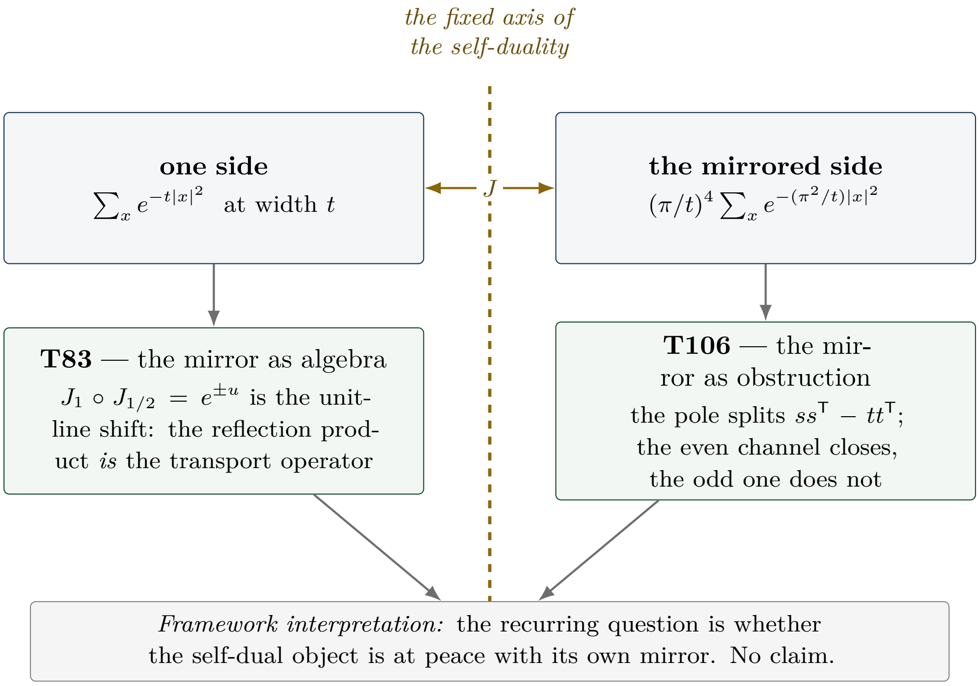

Figure 2: *Schematic.* The self-duality picture. *Literal:* the Poisson identity and its unique self-dual width $t = \pi$ (25 digits, T12); the algebraic identity $J_1 \circ J_{1/2} = e^{\pm u}$ and $\mathrm{Fix}(J_1) \cap \{\text{Weil cone}\} = \{0\}$ (T83); the exact pole splitting $ss^{\mathsf{T}} - tt^{\mathsf{T}}$ at 10^{-18} (T106). *Schematic:* the geometric arrangement, and the reading of the axis as "order" with the lattice's arithmetic content beside it. The bottom line is an interpretation and carries no status.

4 The verified layer: seven modules

This section is entirely [ledger]. Each subsection states the claim as the ledger states it, cites the module, and quotes the ledger's own status markers. Nothing here depends on Sections 5–7.

The seven modules and their ledger markers

Module	Claim id	Content	Markers
v535	HECKE.GEOM.01	Hecke from geometry on the E_8 census channels	[E] + [C] + [O]
v536	HECKE.GEOM.EICHLER.01	Eichler trace layer at the E_8 lattice	[E] + [C] + [O]
v537	HECKE.GEOM.HALFINT.01	half-integral bridge	[E] + [C]
v538	HECKE.GEOM.RTF.01	compiler relative trace identity	[E] + [C]
v539	RTF.GNS.WEIL.01	Weil structure of the compiler family	[E] + [C]
v540	RTF.GNS.AMP.01	amplitude route and positive linear carrier	[E] + [C]
v541	RTF.GNS.LEDGER.01	matching lemma and transport ledger package	[E] + [C]

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Every row carries `NO marker moves` in the ledger. The [O] on v535 and v536 is the same open gate throughout: *weight-4* $\to \mathrm{GL}(1)$ *transport*.

4.1 Hecke from geometry (v535)

The Hecke structure of the μ_4-glue census is lattice-native: it comes out of Kneser p-neighbour geometry, not out of modular input [`verification/v535_hecke_from_geometry.py`] (25 checks, $\sim$11 s; consolidated promotion of T27/T31/T32).

- **Kneser carries Hecke** [E]: $\#\text{iso\_pts} = p^7 + p^4 - p^3$ and $\#\text{iso\_lines} = \sigma_3(p) \cdot \#\mathbb{P}^3(\mathbb{F}_p)$ identically and by full enumeration at $p = 2, 3, 5, 7$ (135/1120/19656/137600 lines). The Eisenstein eigenvalue $\sigma_3(p) = 1 + p^3$ is an exact factor of a lattice-native correspondence count.
- **The marked neighbour sum is an affine Hecke element** [E]: $\nu_p = a\,\mathrm{Id} + b\,T_p$ exactly (overdetermined, residual 0), with $b = \sigma_3 + a_p$; the cusp rule $a_p = b - \sigma_3$ recovers $a_3 = -4$ and $a_5 = -2$.
- **Census redundancy is 2-adic oldform structure** [E]: $\dim V = 7 = 5{+}2$, $\pi_{\text{cusp}} = (28 - T_3)/32$, only 2-adic levels occur.
- Fences carried *inside* the claim: the $p = 5$ geometric $S[1]$ block is a T31-validated *freeze* [C]; the weight-4 $\to \mathrm{GL}(1)$ transport is [O].

4.2 The Eichler trace layer (v536)

The smooth (Eisenstein) background and the coherent (cusp) interference separate exactly [`verification/v536_eichler_trace_layer.py`] (23 checks, $\sim$30 s; promotion of T33/T36/T37/T42). The Witt layer gives $\lambda_{\text{Eis}} = \sigma_3(\#\mathbb{P}^3 - \sigma_3) = L - \sigma_3^2$ identically (anchors 336, 3780, 19264 at $p = 3, 5, 7$) [E], and the Eichler identity $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$ holds two-sidedly (352/3784/19840). The signed cusp reading is $a_p = -c(p)/8 \Rightarrow (-4, -2, +24)$, and the assembler $R(p) = a_p^2$ is exact for $p \le 31$. The $p = 5$ type-(ii) block is again a validated freeze [C]; the transport gate stays [O].

4.3 The half-integral bridge (v537)

In the 70-element compiler weight-5/2 theta monoid there is *exactly one* object g with $\mathrm{Sh}_{t=2}(g) = -8f_8$ coefficientwise [E], it is an exact $T(p^2)$-eigenform with the weight-4 eigenvalues $(-4, -2, 24, -44, 22)$ [E], and $U_4(g) = 0$ places it outside the Kohnen plus space — a scope fence, not a defect [E] [`verification/v537_halfintegral_bridge.py`] (20 checks, $\sim$90 s; promotion of T38/T41/T44/T45). The Waldspurger/Baruch–Mao constant $R = 23.1873585645\ldots$ is constant on ten discriminants $d \equiv 1 \bmod 8$ (spread $\le 10^{-12}$), typed [C] because it is an AFE-numeric measurement.

4.4 One finite relative trace identity (v538)

The strongest structural statement of the verified layer: v535, v536 and v537 are three *projections* of one finite relative-trace identity of Jacquet type [`verification/v538_relative_trace_identity.py`] (18 checks, $\sim$14 s; promotion of T53). $\mathrm{Tr}_V(\nu_p \circ \pi)$ is computed twice, independently — geometrically from the glue census with no modular input, and spectrally from $\eta(2\tau)^4\eta(4\tau)^4$ with no geometry — and the two agree exactly for all $(p, \pi) \in \{3, 5, 7\} \times \{\mathrm{Id}, \pi_{\mathrm{Eis}}, \pi_{\mathrm{cusp}}, \pi_{E_4}, \pi_{f_8}\}$ (anchors, e.g. $p = 7$: full trace 727680, f_8 channel 19840) **[E]**. The Eichler split *is* the orbit decomposition: principal orbit $a\,\mathrm{Id} \to \lambda_{\mathrm{Eis}}$, elliptic orbit $b\,T_p \to a_p^2$, cross terms cancel exactly. And the period side is literally compiler geometry: g is the quaternary lattice form $n = (x^2 + y^2)/2 + 2z^2 + u^2 + 2w^2$ with sign $(-1)^{|u|+|w|}$, giving [signed lattice count]$^2 = R\,|d|^{3/2} L(f_8 \times \chi_d, 2)$ at relative 10^{-12} **[C]**. The identity is *finite*; the infinite RTF is explicitly open.

4.5 The Weil structure of the family (v539)

The family carries a fully identified Weil structure *up to two explicitly isolated obstructions*, and the obstructions are promoted content, not a footnote [`verification/v539_weil_structure_family.py`] (25 checks, $\sim$6 s; promotion of T55/T61/T62/T63). The canonical GNS pairing is a direct integral over the Gelfand spectrum with fibre mixing $\leq 5.04 \times 10^{-16}$; the trivial Sato–Tate isotype is algebraically a GL(1) core, $G_0 = (1 + Y)/(1 - Y)$; the prime side is an exact linear relation to shifted ζ-Weil forms at relative $\leq 9 \times 10^{-16}$. The two obstructions are (i) a sign/doubling residue (metaplectic, sym^2) that survives every preregistered canonical projection, and (ii) a Plancherel correction which is irreducibly non-automorphic. Finite-class positivity of Q_{fam} is measured **[C]** ([4.369, 11.486]); dense-class positivity is open. The ledger row says it in one line: *not almost-RH*.

4.6 The amplitude route and the positive linear carrier (v540)

The route out of the square plane [`verification/v540_amplitude_linear_carrier.py`] (34 checks, $\sim$3 s; promotion of T67–T72). An amplitude Dirac operator $D = \begin{pmatrix} 0 & V \\ V^{\mathsf{T}} & 0 \end{pmatrix}$ with $V[d, m] = \sqrt{w_d}\, b(d)\chi_d(m)$ exists and is Hecke-equivariant **[E]**; the geometric polarisation $b = N_+ - N_-$, $\Theta = N_+ + N_-$ is lattice-exact with $N_\pm \geq 0$ for all $n \leq 50\,000$ **[E]**; the Cohen seed $\Theta(d) = -48L(-1, \chi_d)$ is exact-rational on 159 live d **[E]**; the prime side is a *pure plus* combination $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$ at relative 2×10^{-15} **[E]**; the functional equation of the linear family holds at relative 1.4×10^{-40} **[E]**. The open boundary is *inside* the claim: the gap functional λ^* on $n \equiv 6 \bmod 8$, with an exact Farkas witness (0.0511), is a named limit, and positivity is asserted only in the Euler region.

4.7 The matching lemma and the transport ledger (v541)

The largest module of the line [`verification/v541_matching_lemma_ledger.py`] (33 checks, $\sim$10 s; promotion of T78–T82 and T85; every core check recomputed, not cited).

- **Window proof [E]**: the matching lemma is proved exact-integer on $[4, 10^6]$ — $7S(j) < 40A(j)$ at every atom, 0 violations over 939 870 clash atoms, no-overflow proven in exact integers, 268 big-integer rechecks; exact margin $X = 22242575693/270725400000 = 0.082159$ at $j^* = 554400$; $\rho_{\mathrm{crit}} = 1.143989 > 21/20$. A maximal-greedy identity makes the certificate uniform in the target set and the greedy weights.[1]
- **Transport ledger [E]**: $Q_{\mathrm{Weil}} = Q_{\mathrm{cert}} + \Delta_{\mathrm{arch}} + \Delta_2$ closes with residual 4.4×10^{-16} on a 24-row battery, and $\Delta_{\mathrm{pole}} \equiv \Delta_{\mathrm{conv}} \equiv 0$ are *proven*, not measured (kernel collapse in `sympy`, exact scale invariance).

[1] Prior-art note (literature pass, 2026-07-27): the coefficient laws L1–L4 that T77/T78 use to shape the lemma are specialisations of Cohen 1975 (Math. Ann. 217: the Hurwitz class-number series $H(r, n)$ and the Cohen–Eisenstein series of weight $r + \frac{1}{2}$) and of Pei–Wang 2003 — corollaries of known theory, not discoveries. The restricted divisor inequality itself remains the independent part.

- **Character-exact envelope** [E]: $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8})\Theta$ coefficient-exact for all odd $n \le 50\,000$; confinement holds as a *set equality* on 10^6.
- **Archimedean term internal** [E]: the Legendre duplication bridge $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$ and the kernel identity $k_\zeta = K_{\text{fam}} - K_{\text{shift}}$ at 7.9×10^{-31} pointwise.
- **Frontier facts** [C]: unlifted coherent chain crossing at $k^* = 14$ ($\log_{10} N^* = 23.1$); Robin tail factor 6.16 at $J = 10^6$ and divergent — the constant route stays open.
- **Named limits as content.** Two limits are part of the claim rather than hidden by it: one classically-shaped open lemma (correlated cancellation on credit-rich non-coherent tail atoms), and the I5 equivalence typing discussed next. The row states explicitly: *NOT "almost RH"*; `ZETA.HP.CARRIER` untouched.

5 The localization program (sandbox)

Provenance

Everything from here to Section 9 is [sandbox]: exploratory probes in `experiments/tfpt-discovery/`, no ledger row, no marker, no authority. The findings are reported because the *shape* of the residual is the useful output, not because any of them is established.

5.1 I5: the ledger closes and one inequality is left (T79)

The transport wall — the thing that had blocked the line since T20 — was fully decomposed in T79 (`transport_ledger_probe.py`, 22/22). The key observation is that the odd-prime side of the Weil functional agrees *exactly* with the certified plus combination (relative 2.6×10^{-16}): the prime side was never the wall. What remains after the ledger closes is a minimal transport set of two exact identities (proved), two classically-provable bounds, and exactly one coupled inequality:

$$\textbf{(I5)} \qquad Q_{\text{cert}}(h) + \Delta_2(h) + A_{\text{fam}}(h) - A_{\text{shift}}(h) \;\ge\; 0 \quad \text{for all autocorrelations } h.$$

I5 is the prime ↔ archimedean coupling. By the closed ledger it is, as an identity, equivalent to Weil positivity and hence to RH. The diary is explicit about what that means: *the built-in safeguard fired.* RH does not follow from finite bookkeeping; the hidden infinite step is exactly I5 — and the value of T79 is that the step was *found and named* rather than concealed. This is an equivalence typing, not progress.

T82 then changed I5's *type* (`heat_arch_probe.py`, 22/22, ARCH-INTERNAL): Δ_{arch} is exactly the internal Γ difference $A_{\text{fam}} - A_{\text{shift}}$ via Legendre duplication, verified on 18/18 battery rows at max relative 6.6×10^{-15}. I5 stopped being a coupling between two worlds and became a *self-consistency statement about one heat family.* Again: a type change is not a proof.

T83 (`fe_symmetrization_probe.py`, 27/27, INVARIANT-NULL) then asked the obvious question of the self-duality picture (Section 3): can the functional equation itself do the work? It cannot, and the way it fails is informative. The ζ reflection $J_{1/2}$ acts as the identity on autocorrelations — the symmetry is already absorbed in evenness, and $Q_{\text{Weil}}(Jh) = Q_{\text{Weil}}(h)$ holds floating-point-exactly on 15 test functions — and the family reflection J_1 moves nothing in the value geography (5/24 overlap invariant, λ^* covariant-exact) while pushing m_+h out of the Weil cone in closed form, with $\text{Fix}(J_1) \cap \{\text{Weil cone}\} = \{0\}$ exactly. The structural finding underneath the null is the one quoted in the big picture: the product of the two reflections is the unit-line shift, i.e. the transport operator, and the wall is transversal to the functional equation rather than positioned by it. Symmetrisation is therefore not a route; the same mirror returns, as an obstruction, in T106.

5.2 The Connes dictionary (T87) and where positivity is anchored

T87 (`i5_connes_dictionary_probe.py`, 22/22, DICTIONARY-EXACT-CORE) identified the I5 one-family form as a compiler instance of the semi-local Connes structure. Four exact matches: the two-shell orbit route equals Connes' expression at relative 6.3×10^{-16}; the orbit kernel of Connes–Consani (2021) is literally the compiler's Poisson ladder at $\rho = e^u$ (`sympy` plus 1.7×10^{-50}); h_+ is by definition the compiler kernel $k_\zeta = K_{\text{fam}} - K_{\text{shift}}$; and $k_\zeta = 2\theta'_{\text{RS}}$ — the derivative of the *Riemann–Siegel phase*, at relative 0.0. The dihedral shift is the scaling group $\mathbb{R}^*_+$.

The most useful part of T87 is not a match but a *boundary.* The two positivity programs are complementary: Connes–Consani anchor positivity *before* the primes (the prime-free Sonin window, support $(\frac{1}{2}, 2)$, i.e. $u \in (-\log 2, \log 2)$), the compiler anchors it *after* the primes (atom certificates, all supports). The window boundary is exactly the first prime atom $u = \log 2$ (`sympy`-exact; on narrow rows the prime side, Q_{cert} and Δ_2 all vanish identically). The sectors *touch* and do not overlap; the I5 coupling lives in the crossing region between them, outside both proved sectors.

5.3 The tightness map (T88) and the crossing (T89–T91)

T88 (`i5_tightness_probe.py`, 27/27) mapped where I5 is actually tight: a band-limited zero plateau (classical), safe zones, and *eight* near-vertical tight curves whose spacing follows the Γ-side density law (ratio 1.01 ± 0.08, computed zero-free — no spectral identification is claimed or made). There are no negatives on genuine autocorrelations; the earlier T76 negatives were the shadow of missing archimedean/$p = 2$ bookkeeping, exactly as the ledger predicted.

T89 (`sonin_crossing_probe.py`, 19/19) established that the window compression of the Weil form *is* Bombieri's classical object, and settled the sharpness question: the boundary at $\log 2$ is *not* sharp. The margin falls smoothly from $a \approx 0.5$ towards the floor with no structure at $\log 2$; the prime-free sector still has margin 0.557 at its own boundary and keeps ≥ 0.25 of the reference comfort to $a \approx 0.75$ (+0.057 beyond the proved boundary). The residual subspace grows softly $0 \to 1 \to 2 \to 3 \to 4$ (out of 32) and carries the global minimum. T90 (`residual_dissection_probe.py`, 17/17) made those residual vectors explicit — n-stable to $\leq 2.1°$, closed Gaussian$\times$ cos/Gaussian$\times$ sin fits with 99+% capture — and decided the core question: the residual is *not* just pole coupling (3/10 vectors are controlled by no structure at all; beyond $a \approx 0.85$ genuinely pole-free atom$\leftrightarrow$arch content remains).

T91 (`thin_band_analytic_probe.py`, 19/19) identified the band as the *one-atom zone* and found the mechanism that dominates the rest of the document: beyond an inner edge the prime atom $u = \log 2$ becomes *load-bearing* — the prime *rescues* the form where the archimedean margin is exhausted (rescue term $+0.22 \ldots + 0.41$, purely odd sector). Two uncertainty constants were decided in closed form: $a \cdot t_{\text{rms}} \to \pi$ (Wirtinger, 0.3%) and $a \cdot t_{\text{cent}} \to 2\,\text{Si}(\pi) - 4/\pi = 2.4306$ (two independent routes, 10^{-25}).

The band that these three parts map out is narrow and its landmarks are explicit. Reading outwards from the classical side, the map is: classical edge 0.34657, prime-free crossing 0.3723–0.3763, band top $\alpha^* = 0.46265$, second atom $\log 3/2 = 0.54931$. Everything in Sections 6 and 7 happens inside that interval, between the first and the second prime atom, and the four numbers are the coordinates used throughout. The map was re-derived by a third independent implementation (T93, `band_reconciliation_probe.py`, 41/41) before it was used further.

An attempt to convert the band into a proof was made and did not succeed. T92 (`zone_extension_proof_probe.py`, 36/36, T-SKELETON) certified something small and real — $Q \geq 6.7 \times 10^{-12} > 0$ on an 8-dimensional sine-window subspace across the whole band, with an error certificate and a 1057-point covering — but not the lattice zone extension it was aiming at, and it named two independent walls: $\lambda_{\min}$ collapses geometrically (factor 11.4 per mode), and the high-mode complement would need $N^* \gtrsim 5069$ modes. The finding, and the reason it is quoted here rather than buried, is a statement about instruments: *the finite-block route is structurally the wrong instrument for this target, not an under-resourced one.* That is what motivated the change of approach in Section 6.

5.4 The blind prime demonstration (T94)

A separate, deliberately modest exercise. T94 (`prime_blind_demo_probe.py`, 16/16, BLIND-100) used the exact geometric criterion "odd n is prime $\iff r_4(n) = 8(n+1)$" (Jacobi 1834, on the $\mathbb{Z}^4$ = rank-$|\mu_4|$ theta tower) to predict all 753 primes of the never-seen window [1 000 001, 1 010 000] — count and MD5 declared *before* any ground truth, then evaluated: exactly 753, zero false positives, zero false negatives, 100.00% on all 10^4 integers, with an AST-enforced prediction path containing no divisibility test (no `isprime`, no sieve, no `%`, `//` or `/`). The cost is part of the result: a sieve wins by $\sim$820× (1 ms against 0.83 s). This is structure completeness, not an algorithmic advance — and the *spectral* prediction (zeros → primes) remains bound to I5/RH and is not claimed.

6 The handover induction (T95–T101)

The band work left a picture: each prime atom rescues its own zone and hands over to the next. T95–T101 turned that picture into an induction, and then measured whether the induction can run.

6.1 The relay is real (T95, T96)

T95 (`continuum_extension_probe.py`, 28/28) proved the C1 chain unconditionally ($|h_f(\log 2)| \le 1/2$ by disjoint supports with margin 0.2305; the windowed symmetrised shift has $\|S\| = 1/2$ exactly, 1.1×10^{-16}) and found that the binding mechanism is *not* the two-bump extremal (share 0.046) but the atom rescue: above $\alpha_{\text{free}} = 0.371654$ the prime-free minimiser has $h(\log 2) < 0$ down to -0.328, so the atom adds positivity exactly in the losing direction.

T96 (`relay_test_probe.py`, 21/21) is the model of a self-correcting run. The apparent "edge" of T95 at α^* was a map artefact: the value $+7.8 \times 10^{-6}$ reproduced exactly, but it sits on a smooth, strictly positive curve with no feature at all ($\lambda_{\min} > 0$ on the whole of $[0.38, 0.86]$, decay law $\lambda_\infty \sim \exp(-49.2\alpha)$). The *relay*, however, is real — and it lives entirely in the counterfactual: remove the $\log 3$ atom and $\lambda_{\min}$ crashes to -0.445, with the loser being exactly the anti-double-bump at distance $\log 3$ (alignment -0.99) and a rescue identity at 5×10^{-15}. All four handover windows are positive: each atom arrives strictly *before* it is needed ($+0.025/+0.009/+0.011/+0.007$ for $k = 1..4$). Three consequences were recorded: it is a margin problem, not an edge problem; numerics as a witness is exhausted beyond $\alpha \approx 0.55$; and the counterfactual is the proof target, because it works on $O(0.1)$-sized quantities instead of 10^{-6}.

6.2 The induction step and its one missing lemma (T97, T98)

T97 (`relay_induction_probe.py`, 105/105) gave the induction step proof shape. Alignment is true and sharp with a constant law; the $t = 0$ killer loss is proved by three structure identities (old atoms vanish identically on E_-; the pole rescue flips with factor $1 - \cosh(u/2)$; $|\hat f|^2 = 2(1 - \cos(tu))|\hat g|^2$), giving $k_{\text{eff}} = (1 - \cos(tu))k(t)$ with infimum $-1.11\ldots - 2.59$ instead of $k(0) = -5.37$.[2] The structural pearl: E_0 is *literally* the same form at a smaller window (7×10^{-14}) — the induction hypothesis, by self-similarity. What remained was a single cross-block lemma about a single vector.

T98 (`cross_lemma_probe.py`, 44/44) attacked it and produced the most instructive negative result of the series: the measured $\sqrt{\lambda_0}$ law is Douglas (1966) range inclusion, i.e. a *tautology* of the positivity one is trying to prove — the identification *dissolves* the lemma rather than proving it. Three T97 premises were refuted in the same run. The replacement target is an exact scalar inequality with no constants and no near-null vector:

$$D_k(\alpha) \quad := \quad -\lambda_{\min}\Big(Q|_{E_-} - Q_{-0}(Q|_{E_0})^{-1}Q_{0-}\Big) \quad \le \quad \tfrac{\mu_k}{2}.$$

[2] In closed form (a T92 side result) $k(0) = -\gamma - 3\log 2 - \pi/2 - \log \pi$. Prior-art note (literature pass, 2026-07-27): in substance this is the classical digamma value $\psi(1/4) = -\pi/2 - 3\log 2 - \gamma$, shifted by $-\log \pi$, as it appears in Suzuki 2023 (J. London Math. Soc. 107) — a good consistency check, not a find.

T99 (`dk_recursion_probe.py`, 23/23) then found the a-priori structure: an exact mirror-parity selection rule ($J_- Q_{-0} J_0 = -Q_{-0}$ at 2×10^{-15}) giving $D_k = \max(D_{\text{even}}, D_{\text{odd}})$, with the binding even channel coupling only to the *odd* spectrum of the smaller window — so the fragile near-null mode is identically absent from the binding channel. T100 (`bulk_remainder_probe.py`, 27/27) closed the bulk remainder from 11/24 to 24/24 samples and showed the 10.8% drift was a grid artefact.

6.3 The race (T101)

T101 (`k_asymptotics_probe.py`, 31/31, CROWDING-TRENDS) resolved 16 zones ($n = 2 \ldots 29$) and separated two questions that had been fused. The *mathematics* trends self-supportingly: the primitive quantities are flat over 16 zones (w/g slope $+0.002 \pm 0.178$; the relative D_k margin does not trend; $D_k \leq \mu_k/2$ holds 64/64 and never fails). Atom *strengths* do not fall ($\mu \sim n^{-0.087}$), only the *spacings* do ($n^{-0.615}$, exact) — pure arithmetic. What loses the race is the *closure instrument*: the demand ratio r_k decays like $\exp(-0.1622k)$ (fit, ± 0.0562). The collapsed one-parameter law $w_k = 0.0838 \cdot (\text{atom spacing})/\mu_k$ reduced the scatter from 8.85× to 2.66× with zero residual trend.

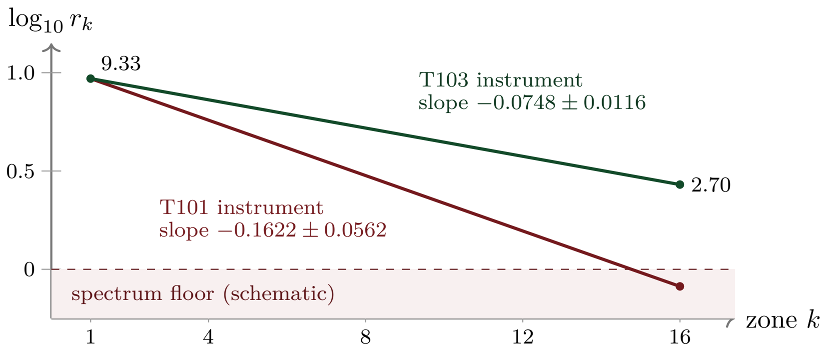

Figure 3: *Schematic.* The instrument race over the 16 resolved zones. *Literal:* the two decay slopes (-0.1622 ± 0.0562 measured in T101; -0.0748 ± 0.0116 measured in T103, a factor 2.2 flatter), and the T103 endpoints $r_1 = 9.33 \to r_{16} = 2.70$ with the statement that the improved instrument never leaves the spectrum. *Schematic:* the straight-line interpolation in $\log_{10}$, the common anchor of the two lines, and the position of the "spectrum floor". Both slopes are declared fits in the diary. What loses the race is the instrument, not the inequality: the true inequality $D_k \leq \mu_k/2$ holds 64/64 and its relative margin does not trend.

T101 also stated what an asymptotic theorem would need, and that list has not changed since: (A) the collapsed lower bound $w_k \geq c\, g_k/\mu_k$ with a fixed $c > 0$ — a sharp, non-perturbative, *arithmetic* lower bound at a Weil-positivity edge, with Taylor arguments forbidden by the $\exp(-c/\delta')$ onset; (B) a uniform relative margin; (C) a replacement for the bulk remainder; (D) a finite check of the low zones. If crowding ever wins, the hardness sits entirely in (A).

7 The certification arc (T102–T118)

The last seventeen parts stopped improving measurements and started converting them into certificates — and, where that failed, into explicit refutations.

7.1 The mechanism: a crossing, not a singularity (T102)

T102 (`arithmetic_bound_probe.py`, 42/42, MECHANISM-IDENTIFIED) computed the mechanism exactly. The k-th atom acts on $E_-/E_0/E_+$ as $\operatorname{diag}(-\frac{1}{2}, 0, +\frac{1}{2})$, so $Q_{k-1} = Q_{\text{full}} - \frac{\mu_k}{2} P_- + \frac{\mu_k}{2} P_+$: the atom strength μ_k enters exactly once, linearly. That gives a two-sided sandwich through the

Schur profile $\sigma_k(\delta)$, with $\mu_k/2 \le \sigma_k$ implying that the handover holds, and $2w_k$ lying between the crossings in 16/16 zones.

Two corrections came out of the same run. First, the onset is *anchored* at $\delta_c = 2w_k$ — a crossing of two finite quantities ($R^2 = 0.968$) — and exponential-singular versus power behaviour is not separable; T96's essential-singularity reading is compatible but no longer distinguished. Second, and more consequentially, the binding constraint *flips*: no concentration condition binds. The bare E_- form is strongly positive ($2.65 \ldots 3.52$, i.e. 4–14× $\mu_k/2$), the pole flip and band mass saturate their classical ceilings (96.9–97.1% of Cauchy–Schwarz and Landau–Pollak/prolate respectively), and the onset is manufactured *entirely* by the Schur dressing against $E_0 \oplus E_+$, which eats 35.7%–97.3% of the bare eigenvalue. The loser does not live in E_-.

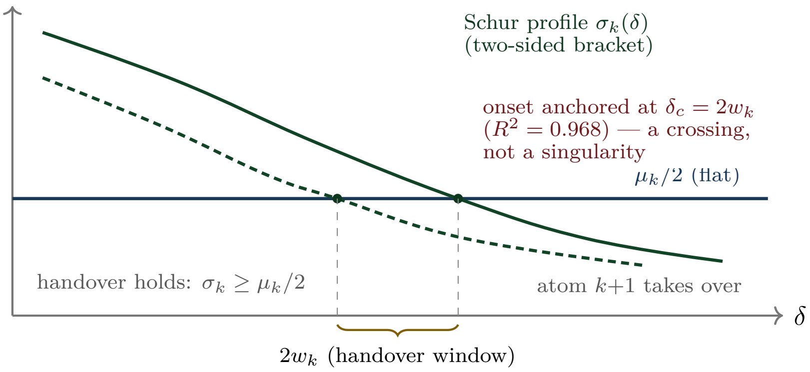

Figure 4: *Schematic.* The crossing mechanism of T102. *Literal:* μ_k enters exactly once and linearly, so the condition is a two-sided sandwich through the Schur profile $\sigma_k(\delta)$ against the flat level $\mu_k/2$; the window width $2w_k$ lies between the crossings in 16/16 zones; the onset is anchored at $\delta_c = 2w_k$ with $R^2 = 0.968$; the candidate bound $w_k \ge A_k \mu_k^{1/q_k}$ holds 16/16 with ratio 0.749–0.940. *Schematic:* the curve shapes and all absolute scales.

The same run performed the most useful kill of the arc: a triple decomposition test showed that the T101 law in C/g was a *proxy*. g_k is causally impossible (the form Q_{k-1} is blind to the next atom), statistically dispensable (a causal law $C \cdot u$ has CV 23.3% against 27.6%), and arithmetically only a ceiling ($C_k \le g_k \mu_k$ exactly; the 16-zone extrapolation violates the ceiling from $k = 69$; checked over 18 120 prime-power atoms to $n = 200\,000$).

7.2 The instrument, rebuilt (T103), and the two doors

T103 (`instrument_probe.py`, 29/29, INSTRUMENT-IMPROVED) rebuilt the closure instrument with certified weights $w_j \ge 1/\lambda_j$ and a θ-weighted band sum, reproducing both anchors exactly. With *one* fixed, k-uniform instrument ($\Lambda_0 = 3$, $r = 2$, no per-zone tuning) the closure map jumps from 7/64 to 44/64 and the race slope halves (Figure 3). The price is explicit: the number of explicit modes grows $2 \to 232$. And the verdict relocated the remainder: it is no longer the bulk (the bulk is measurably not low-rank — effective rank up to $0.579 \dim E_-$) but the *wing slack* $S = 1 - \rho$, which falls $0.2091 \to 0.0392$.

The convergence recorded at the end of T102 is the pivot of the whole arc: T103's wing slack $S = 1 - \rho$ and T102's Schur dressing are *the same object seen from two sides*. Both doors lead to the wing near-null direction.

7.3 Inductive chains, and the death of the margin route (T104)

T104 ran as *two independent arms* after a file collision between two parallel workers was resolved by keeping both implementations of the same contract — `schur_profile_bound_probe.py` (21/21) and `schur_profile_chain_probe.py` (47/47). All core findings converged, which makes it a de-facto independent replication.

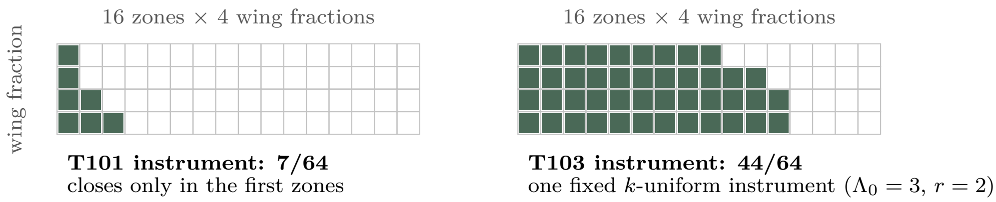

Figure 5: *Schematic.* The closure map. *Literal:* the grid is 16 zones × 4 wing fractions = 64 cells; the old instrument closes 7 of them, the rebuilt one closes 44 with a single fixed k-uniform setting ($\Lambda_0 = 3$, $r = 2$) and no per-zone tuning. *Schematic: which* cells are shaded — the diary records the totals, not the pattern; the shading is drawn as a monotone wedge purely to visualise the counts.

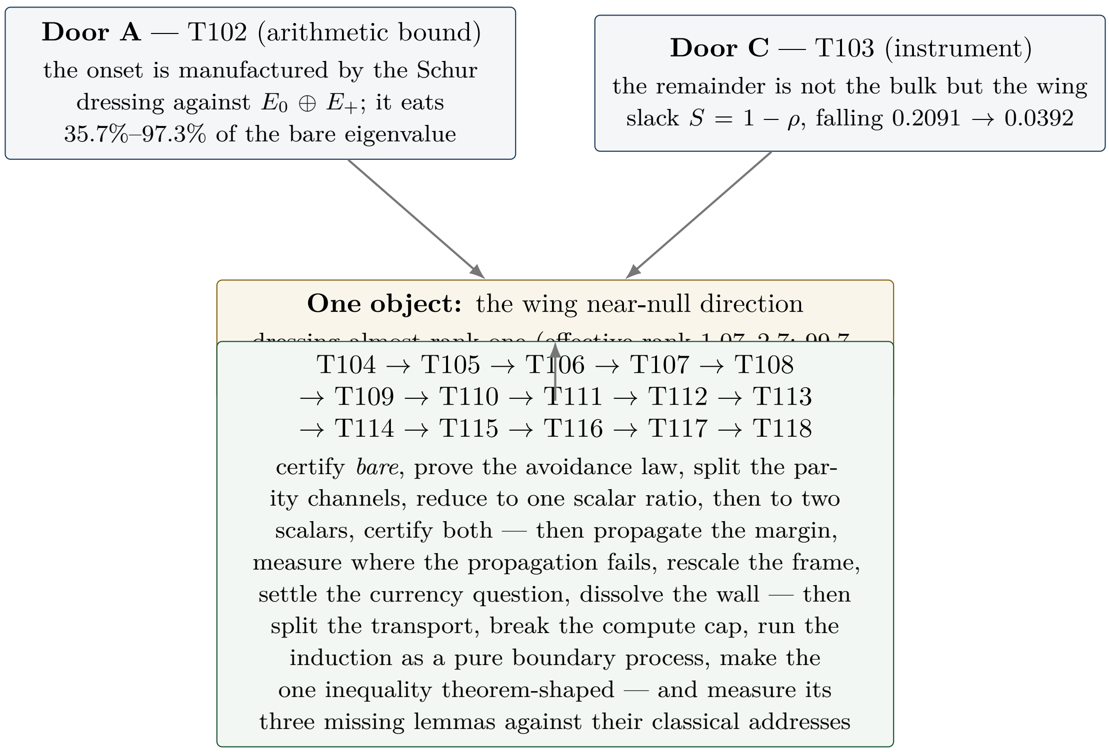

Figure 6: Two doors, one object. Both entries are literal diary findings; the convergence statement (“the same object from two sides”) is quoted from the T102 closing note.

- **The naive margin route is dead**: $M \geq m \cdot \mathrm{Id}$ holds in 0/16 zones; the coupling mass to margin ratio is 10^3–10^6; m vanishes in the continuum like $M^{-1.7}$ (fit).
- **The mechanism is avoidance**: the coupling B_- decouples from the soft bulk modes — the softest mode carries 0.00% of the coupling mass for $n \geq 3$, and the mass sits in mid/high bands (lowest band only 2–12%). Positivity is not saved by a margin but by structural avoidance.
- **Exact spectral-split chains close** 16/16 with finite data: arm A via the exact block-inverse identity with matrix caps in PSD order (headroom 1.15–1.44, reach $\gamma_{\max} = 0.50$–0.90); arm B via a threshold split with an $O(1)$, resolution-stable threshold $w = 2.47$–5.90 — but requiring $r_{\min} = 64 \ldots 1024$ soft modes $\propto M$, which makes the continuum statement a spectral-density condition.
- A recommendation from T96/T103 was tested and *rejected*: the prolate wing basis loses against the raw basis in both arms.

Consequence: the hard core migrated from σ_k as a whole to four named ingredients — a lower bound on $\mathrm{bare}_k = \lambda_{\min}(Q_{\mathrm{full}}|_{E_-})$, the soft dressing scalar L, finite induction data, and a one-sided edge estimate instead of a fit. The chain around them is classical (Schur complement, exact block inverse, Parseval/Bessel, Weyl) and fit-free.

7.4 The certified bare bound and the avoidance theorem (T105)

T105 (`bare_wing_bound_probe.py`, 28/28, ONE-OF-TWO) delivered one of the two missing ingredients in closed form. First it resolved the T104 arm discrepancy: *bare* depends only on (u_k, δ) — the window centre cancels ($\le 0.74\%$ drift under $8\times$ refinement) — with the exact split bare $= \mu_k/2 + b_0$. Then three classical steps (Bessel; a Legendre/level-set optimisation at $k(\pi/\delta)$; Cauchy–Schwarz at the pole pair) collapse to a closed lower bound:

$$b_0 \quad \ge \quad \frac{\delta}{\pi}\int_0^{\pi/\delta} k(t)\,dt \quad - \quad \delta\,K_{\max} \quad - \quad 4\left(\cosh\tfrac{u}{2} - 1\right)\sinh\tfrac{\delta}{2}.$$

This is the band average of the archimedean kernel over the band conjugate to the wing, of size $\sim\log(1/2\delta) - 1$. It is positive in 16/16 zones (1.27–3.54) at 81.1–92.7% of the measured value (median 92%), it sharpens as the wing flattens, and it needs no eigenvalue or induction data.

Second, the *avoidance law* became a theorem rather than an observation: $Q_{\text{full}} \succeq 0 \Rightarrow \|B^\mathsf{T} v_i\|^2 \le w_i\Lambda_-$ (a semi-inner-product argument; coupling proportional to the mode eigenvalue), verified on every mode of every zone with worst ratio 0.377. That explains the measured 0.00% avoidance exactly. Alongside it, an exact parity superselection: $JQJ = Q$ (5.7×10^{-15}) and $JB = -BR$ (3.4×10^{-15}) — the two channels do not mix, and the softest mode is J-even in 16/16 zones. Notably the *Euclidean* angles are not small (cosine up to 1.0): the avoidance lives in the form metric, as a Friedrichs angle.

What T105 left is one Loewner statement: $Q_{\text{full}}(\alpha) \succeq (\mu_k/2)P_-^{(k)} \iff \rho \le 1 - (\mu_k/2)/\text{bare}_k$ — a Friedrichs-angle / spectral-density condition, explicitly *not* a margin condition. The free cap $L \le \Lambda_- = 5.63$–7.52 diverges like $\log(1/D)$, so the certificate has to come from the spectral density near the soft end.

7.5 Parity superselection: the pole splits (T106)

T106 (`friedrichs_angle_probe.py`, 32/32, DENSITY-MAPPED) killed two routes and found the structural result of the arc. The kills are in Section 8. The finding:

$$ab^\mathsf{T} \;+\; ba^\mathsf{T} \qquad = \qquad ss^\mathsf{T} \;-\; tt^\mathsf{T} \qquad (10^{-18};\ Ja \;=\; b)$$

Here J is the mirror parity of Section 3, arriving for the second time and now as the obstruction itself. The rank-2 Weil pole splits *exactly* into a positive rank-one lift in the J-even channel and a negative rank-one pressure in the J-odd channel. The Toeplitz part has exactly one negative eigendirection; it is J-even, and ss^T removes it precisely there. The odd channel never needed the pole at all.

The channel-wise angle chain closes 16/16 in the even channel and 9/16 in the odd channel, against only 3/16 unsplit; all 16 handovers close at operating depth in both channels under the exact Schur criterion, with certified b_0 at 89–93%. Certified after T106: bare, the parity split, the pole splitting, the channel inertia, the avoidance law, the growth inequality, the accumulation bounds. Measured: ρ_-.

7.6 One scalar ratio (T107)

T107 (`odd_channel_closure_probe.py`, 30/30, SCALAR-TRACTABLE) reduced the matrix statement (R) to a scalar one. Sherman–Morrison gives exactly (R) $\iff s^* \le 1$ (identity at 3.4×10^{-11}; equivalence confirmed 16/16 against Rayleigh–Ritz; precondition $G \succ 0$ via Cholesky 16/16) — but s^* is *saturated* ($0.99885\ldots0.99999$, with only $n = 2$ at 0.2569), so no uniform s^* chain can work. The Woodbury decomposition rescues it: $s^* = \tau + \kappa$ and

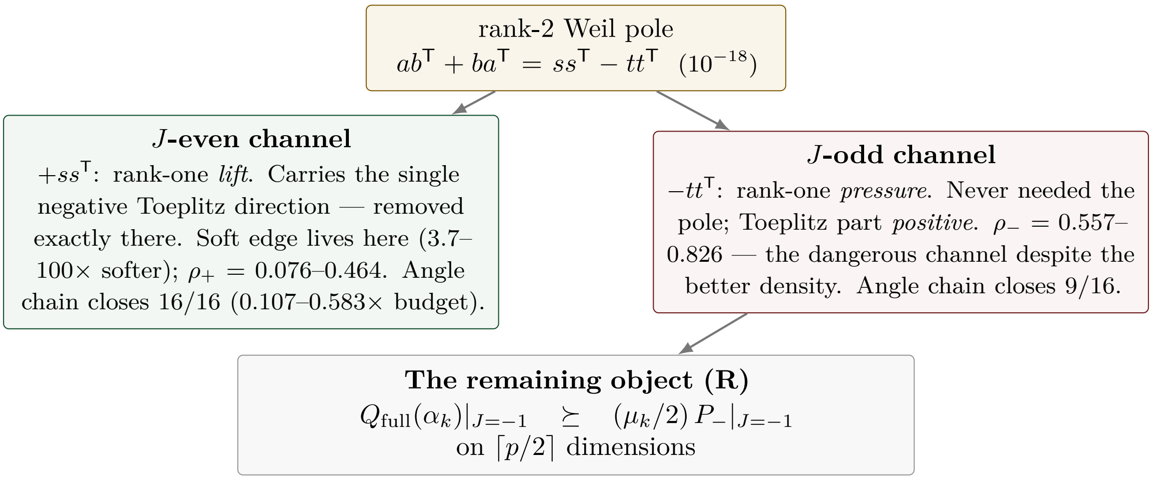

Figure 7: Parity superselection (T106). All numbers are literal; the layout is illustrative. The practical consequence: it is not the density that decides the angle but the *avoidance* — the dangerous channel is the one with the better density. T105's certified bare bound sits 16/16 exactly in the hard channel, and ρ_- is resolution-stable ($\leq 17.6\%$).

$$\text{(R)} \quad \Longleftrightarrow \quad r := \frac{\kappa}{\varepsilon} \leq 1, \qquad \varepsilon := 1 - \tau, \qquad \text{measured } r = 0.0053 \ldots 0.1814.$$

Two orders of magnitude of headroom instead of five decimal places, and s^* is resolution-stable (spread $\leq 2.60\%$ over $M = 600/900/1200$). The synthesis is stated with the usual asymmetry: certified-closed 0/16, measured 16/16 (with $q = 1.31$–4.16 at $M = 1200$, replacing T106's 0.747–1.156 — a factor 1.7–2.5 gain). Exactly one object is missing: a certified positive lower bound for $\varepsilon = 1 - \tilde{t}^\mathsf{T} T_{\text{odd}}^{-1} \tilde{t}$ — how far the pole vector is from exhausting the odd channel — or, since ε and κ both fall with M ($\varepsilon \sim M^{-1.99\ldots-1.69}$, fit), a certificate for the *ratio* r itself. The named route is via a wing statement of the T105 type plus an avoidance norm, explicitly not via Szegő.

7.7 The ε identity, and two scalars (T108)

T108 (`ratio_certificate_probe.py`, 44/44, EPSILON-IDENTITY) turned the missing positivity of ε from a bound to be won into an *identity*. Everything hangs on the pole response $x := T_{\text{odd}}^{-1}\tilde{t}$, and three exact identities (verified at 10^{-11}–10^{-13}) pin it down: $\varepsilon = x^\mathsf{T} Q|_{\text{odd}} x/\tau$ — so ε *is* a $Q|_{\text{odd}}$-energy, not an accident of subtraction — $1/\varepsilon = 1 + \tilde{t}^\mathsf{T} Q|_{\text{odd}}^{-1}\tilde{t}$, and the overlap of ε and κ is exactly one rank-one term $hh^\mathsf{T}/\varepsilon$, so that (R) $\Longleftrightarrow$ $\hat{r} = (\mu/2)\lambda_{\max}(\Gamma + hh^\mathsf{T}/\varepsilon) \leq 1$. The third route supplies the classical reading: $\tilde{t}$ is exactly two-term geometry, τ a 2×2 Christoffel–Darboux form, and ε exactly the square of the last Cholesky pivot — the Szegő/Levinson prediction error. *The positivity of ε coincides with the induction positivity in the odd channel*; it is not a second, independent thing to certify.

Two consequences, and one correction of the previous picture. The consequence: (R) reduces from an $M/2$-dimensional Loewner statement to **two scalars** — $\omega < 1$ (measured 0.2366–0.7531, 16/16; the T105 structure survives the parity split verbatim, $V^\mathsf{T} S|_{\text{odd}} V = -\frac{1}{2}I$ at 4.9×10^{-14}) and one statement at the vector x. The chain (R) $\Leftarrow [\omega < 1] \wedge [\lambda_{\min}(Q|_{\text{odd}}) \geq (\mu/2)\tau\theta/(1-\omega)]$ has $C1$ closing 16/16 (margin 5.5–178; ≥ 1 at all 48 ladder points, 2.7–343), $C2$ 15/16, $C3$ 0/16 — the first time a margin chain in this arc is *not vacuous*: the induction margin it needs is explicitly 9.3×10^{-8}–3.7×10^{-3}, i.e. two to six orders of magnitude *below* $\mu/2$. The correction: the soft-edge picture of T107 was wrong in its mechanism — τ sits to 99.66% on modes $\lambda \geq 1$, and $\|h\|^2$ is small by sign cancellation ($\sum |k| = 401$ versus $\sum k = 1$), not by softness. T107's Cauchy–Schwarz chain turns out to be exactly the Weyl bound (slack 1.0000).

What is exactly open after T108 is one boundary value. The remaining object is a certified upper bound for the avoidance norm $\|V^\mathsf{T} T_{\text{odd}}^{-1}\tilde{t}\|^2$, and on 8 zones that is literally the single edge entry x_0 of an explicit vector (suppressed at $|x_0|/\max|x| = 2.9 \times 10^{-3}$–$5.7 \times 10^{-3}$, with a linear edge profile $r^{1.01}$ quoted as a fit) — a boundary-decay statement about an explicit vector, the same object class as the T105 carrier separation. $\hat{r}$ is flat in M ($6.1\times$ more stable than ε) and $r, \hat{r} < 1$ at all 92 (zone, M) points up to $M = 3000$, tightest at 0.9648: measured, on resolved zones, as always. That boundary value, and the still-uncertified ω, are exactly what the next part attacked.

7.8 Boundary certificates: cancellation, not decay (T109)

T109 (`boundary_decay_probe.py`, 29/29, BOUNDARY-CERTIFIED) closed the arc: both remaining scalars are now certified, and the whole chain for (R) closes 16/16 conditional on exactly one explicit input. Three findings and one honest remainder.

First, the mechanism. The edge value x_0 is not small because anything *decays* — it is small because a sum *cancels*. Carrier separation is excluded geometrically: $\tilde{t}$ has its maximum at $r = 0$; the source sits on the boundary itself. The Combes–Thomas hypothesis is satisfied (T_{odd} is of Jaffard class, with local rate $\rho(s) = D(\frac{1}{2} + 2/(e^{2s} - 1))$, matched at a factor 0.85–1.00, quoted as a fit) — but its conclusion is empty: the Green sum for x_0 cancels by a factor 44–4.6×10^4 on 15/16 zones (the known outlier $n = 2$ does not cancel), and the edge profile is a nearly linear zero, $r^{0.77\ldots1.28}$ as a fit. A boundary condition, not a decay length — and no absolute-value bound can survive a cancellation.

Second, the x_0 certificate that works. Four candidates were run certified-against-measured: Combes–Thomas, evaluated with the *exact* Green row so that no constant could rescue it, is *refuted* (1/16; an upper bound on its whole class); a T-metric Cauchy–Schwarz at the edge entry closes 1/16; the exact Szegő/Levinson two-point evaluation exhibits the cancellation but bounds nothing. The fourth is the new object: a residual / goal-oriented certificate (Prager–Synge; second order in the two residuals) which *carries* the cancellation by construction instead of bounding it away. Sharpness 1.0000–2.4937 against the measured norm, 16/16 within budget.

Third, ω is cracked — unconditionally. The certificate is a graded matrix cap in PSD order: one trial level per soft direction, validity by a single Cholesky factorisation (Sylvester inertia) plus Loewner antitonicity of the inverse, $n_{\mathrm{top}} = 17$–512. It gives $\omega_{\mathrm{cert}} = 0.2561$–$0.7569$ against the measured 0.2366–0.7531 — a factor 1.004–1.124 — with *no* hypothesis input. The compression Schur distance that blocked T108 shrinks from 0.18–0.52 to 0.0024–0.039, and the ungraded cap (one level for the whole soft block, which is what T108's global route amounted to) is vacuous everywhere: the grading is the whole difference. The trial data are numerically generated; their *validity* is trial-independent (identity + Cholesky + Loewner), only their sharpness is not.

The synthesis feeds both certificates from one object ($\Gamma_{\mathrm{cert}} = S_{\mathrm{cert}}^{-1}$): the chain closes 16/16 with margin 1.3–10.4 and is M-stable (44/48 ladder points to $M = 3000$; the certification price 1.5–28.6 does not grow). What remains open is exactly **one input**: the *strict* positivity margin of the odd channel, $\lambda_{\min}(Q|_{\mathrm{odd}}) \geq 1.7 \times 10^{-6}$–$7.5 \times 10^{-3}$ against a measured 1.4×10^{-5}–5.1×10^{-2} — two to six orders of magnitude *below* $\mu/2$, i.e. strictly weaker than the conclusion it buys; the induction hypothesis as it stands supplies only the non-strict $Q|_{\mathrm{odd}} \succeq 0$. Declared caveats: floating point is not audited, and $\lambda_{\min}$ is read through Rayleigh–Ritz. Whether that one number propagates itself through the induction is exactly what the next part measured.

7.9 The margin propagates: the circle closes on the measured zones (T110)

T110 (`margin_propagation_probe.py`, contract `MARGIN.PROPAGATION`, 28/28, MARGIN-PROPAGATES — with a precision statement) asked the only question T109 left open: does the strict scalar margin $m_k = \lambda_{\min}(Q|_{\mathrm{odd}})$ propagate itself along the zone ladder, against the explicit demand need109(k) of the T109 chain? Three findings close the circle on the measured zones; three gaps state precisely what a theorem would still need.

The margin map. On a common grid (16 zones, $\delta_0 = 0.00284$) the margin falls monotonically, 6.95×10^{-2} ($n = 2$) to 1.66×10^{-5} ($n = 29$), against need109 $= 3.09 \times 10^{-7}$–3.87×10^{-4}: $m_k \geq$ need109 on 16/16 zones, with ratio 2.09–179.6. Dilution during pure window growth is weak

RH $\iff$ Weil positivity $\iff$
I5 (one-family form; T79/T82, ledger `v541`)

↓ equivalence *typing*

per-zone **handover induction**: $D_k(\alpha) \leq \mu_k/2$ (T98; holds 64/64 on the 16 resolved zones)

↓ architecture of a *sufficient* route, on finite windows

Schur profile sandwich: $\sigma_k(\delta) \geq \mu_k/2$ (T102; μ_k linear, onset anchored at $\delta_c = 2w_k$)

↓

bare$_k$ **certified** + soft dressing L (T105; b_0 closed form, positive 16/16 at 81.1–92.7%)

↓

parity split: even channel closes 16/16; hardness $\to$ odd channel (T106)

The asymptotic wall. Every step below the top line is established on *16 resolved zones* at finite resolution. Sixteen zones are not all zones; no uniform-in-k constant is certified anywhere in this cascade.

↓

(R): $Q_{\text{full}}|_{J=-1} \succeq (\mu_k/2)P_-|_{J=-1}$ on $\lceil p/2 \rceil$ dimensions (T106)

↓ exact reduction

$\iff$ $s^* \leq 1$ (saturated) $\iff$ $r = \kappa/\varepsilon \leq 1$ (T107; measured 0.0053–0.1814)

↓ exact reduction

$\iff$ two scalars: $\omega < 1$ and one statement at $x = T_{\text{odd}}^{-1}\tilde{t}$ (T108; ε identity; boundary value x_0)

↓ two certificates

both scalars certified: $\omega_{\text{cert}} < 1$ unconditional (graded matrix cap) + residual certificate carrying the boundary cancellation (T109; closes 16/16, margin 1.3–10.4)

↓ one input left

open **input**: one strict-positivity margin $\lambda_{\min}(Q|_{\text{odd}}) \geq 1.7 \times 10^{-6}$–$7.5 \times 10^{-3}$ (T109; 10^2–10^6 below $\mu/2$ — the cascade ends here)

Figure 8: The reduction cascade, with its arrow types made explicit. The top line is an *equivalence typing* produced by an exact identity (Section 5); the arrows below are the architecture of a would-be sufficient route, verified on finitely many windows only. Since T109 the cascade ends at one strict-positivity margin input; T110 shows that this input propagates itself along the measured ladder (certified base case, graded Loewner minorant), leaving three gaps — which T111 then drives deep and measures as three separate walls (the two subsections below). The dashed box is the reason this figure is not a proof sketch.

— a factor 1.02–2.05 per atom gap, i.e. 0.993–0.997 per cell, a $1 - O(1/M)$ loss; the scaling fits are honestly poor (the power law wins 7/16). The structural finding of the part is the atom entry: it costs *nothing*. The naive expectation was a $-(\mu/2)$ pressure on the odd channel; measured, the entry *raises* $\lambda_{\min}$ on 15/15 handovers (the first-order Kato term has helping sign), and at $n = 29$ the atom-free window is not even positive (-9.2×10^{-3}): the atom makes the window co-positive. The T106 killer argument — transport destroys a demand that *points* somewhere — has no analogue for the scalar floor.

The step law. The growth step splits exactly (embedding error $\leq 2.6 \times 10^{-14}$), and $\max |\mu N| = 0.0$ on 15/15: the new atom lies outside the old lag reach (gap 0.0606 against $\delta_0 = 0.00284$), so its restriction to the old window is the exact zero matrix — everything is decided by the bordering. Scalar routes fail decisively: bordered Weyl closes 0/15 and the Schur/Friedrichs scalar cap is vacuous 15/15 (both recorded in the kill list). What passes certified is the **graded Loewner minorant** — its validity condition is exactly the induction hypothesis, and the level is then found by bisection through a Cholesky/Sylvester factorisation: with one soft direction ($n_{\text{soft}} = 1$) it holds 15/15 with retention 1.0000; the strictly scalar variant ($n_{\text{soft}} = \dim$) closes 0/15.

Base case and circle test. The base case $n = 2$ is *certified* by an explicit Cholesky of $Q|_{\text{odd}} - \lambda I$ at three grid resolutions ($\lambda_{\min} \geq 6.93 \times 10^{-2}$, a factor 88–180 over need). The circle test then runs the whole chain from that certified base value: the strictly scalar induction breaks at $k = 3$ (its Schur factor compounds to 2.7×10^{-36}), but the graded chain runs through completely — 16/16 zones above need109, each step one Cholesky, with the propagated floor as the only input. On the measured zones, the induction circle closes end to end.

The three gaps — the asymptotic mountain, made precise

1. **No reserve.** f_{crit} at the first handover is 1.00; the chain lives on a retention of $1 - 2 \times 10^{-7}$, and a factor-2 loss anywhere would break it within two steps. What a theorem needs is a step law *without* loss.
2. **No scalar step law.** Structurally excluded, not merely missed: the new cells attach at the boundary layer where the pole source sits, so any one-number summary of the bordering is blind to exactly the direction that decides it.
3. **No uniformity in k.** Every certificate above is one finite Cholesky; flat-in-k is a measurement, not a theorem. And the measured trend warns rather than reassures: $m_k \sim n^{-1.93}$ against need109 $\sim n^{-0.98}$, ratio $\sim n^{-0.96}$ (fit, rms 0.735) — extrapolated crossing at $n \approx 170$. Propagation gets *harder* with k, not easier. (T111 subsequently *measured* that crossing: it sits at $n^* \approx 462$, not 170 — the extrapolation was a fit artefact of the 16-zone window, but the wall itself is real. See the next subsection.)

7.10 The three walls (T111)

T111 (`deep_zone_stress_probe.py`, contract `DEEP.ZONE.STRESS`, 23/23, CROSSING-CONFIRMED) stopped extrapolating and measured. On exactly the T110 cell grid — the first 16 zones reproduce T110 bit-exactly — the ladder was extended to *every* prime-power zone $n \leq 1000$ plus a thin tail to 3001: 199 zones in the margin map, with the chain ladder running $n = 2 \ldots 521$ through 117 handovers.

The margin wall, measured. The ratio $m_k/\text{need109}$ falls $179.59 \to 0.0174$ across the deep ladder, and the crossing is now a measurement, not a fit: the last zone with ratio ≥ 1 is $n = 461$, the first below is $n = 463$ (ratio 0.935). The $n \approx 170$ figure of T110 was a fit artefact of the 16-zone window — the wall sits much further out, but it is real. The failure follows the *prime branch*: prime-power zones (512, 529, 625, 729), whose atoms are weaker, survive longer than the primes around them. The fit families were again compared honestly (best by AIC: a broken-scale law with $n^* = 501$; the jackknife band of the pure power law is $[298, 352]$), and a depth-preserving resolution study shows that refinement pushes the ratio *down*: $n^* \approx 462$ is an *upper* bound, not an estimate to be rounded up.

The ladder wall, arithmetic. The uniformity object is no longer flat: n^*_{soft} runs $1 \ldots 8$ across the

117 handovers. But retention stays 1.000000 at every step (largest single loss 5.96×10^{-8}), and the atom entry stays constructively free on 117/117 — the restriction of the new atom to the old window is the exact zero matrix, deep into the ladder. The new and harder finding is the *ladder wall*: the nested step needs a lag gap exceeding twice the window depth, and the twin-prime pair $521 \to 523$ (gap $0.003831 < 2D = 0.005617$) kills the ladder *arithmetically* — at frozen cell width, $n = 521$ is the deepest reachable zone, and the bound is set by twin primes, not by spectra. An arithmetic scan to 40 000 finds 4052/4284 consecutive prime-power pairs endangered; letting D shrink like $1/n$ instead forces the cell count to grow like $n \log n/4$ (horizon $n \approx 690$ at $h \leq 1500$).

The circle tears at the ratio, not at a step. The no-reserve finding of T110 softens: $f_{\mathrm{crit}} \sim n^{-0.39}$ (fit) — the reserve *opens* with depth, and the bottleneck stays at the first step $2 \to 3$. The circle test runs from the certified base case (factor 179.6) and breaks first at $k = 109$, $n = 463$ — exactly where the margin map measures the crossing. All 117 handovers certify at retention 1.000000, including beyond the break: what fails is the demand ratio, never the handover mechanism.

The three walls — and the reinterpretation

The hardness balance names the driver: the growth of $1/\kappa$ (exponent $+0.86$) against the margin (exponent -0.68). Three separate walls, in order of appearance:

1. **The margin wall**, $n^* \approx 462$: measured, an upper bound (refinement pushes it down); the T109-chain demand ratio crosses 1 between $n = 461$ and $n = 463$.
2. **The ladder wall**, $521 \to 523$: purely arithmetic — at frozen cell width the nested handover dies at the first twin-prime gap below $2D$, independently of any spectral quantity.
3. **The requirement wall**, $n = 727$: $\omega_{\mathrm{cert}} \geq 1$ makes need109 vacuous on 46 deep zones — beyond it the T109 chain no longer states a demand at all.

The reinterpretation that comes with the measurement: the operating variable is the window *depth*, not the zone index n — doubling the cheeks moves the margin wall from 463 to 47. A proof along this route must beat the exponent gap $-0.974 = -0.681 - 0.293$ and couple the depth δ_0 to the local prime gap.

T112 subsequently ran the construction in exactly that coupled frame (next subsection): walls (2) and (3) fall *structurally*, wall (1) proves frame-invariant.

7.11 The scaled frame: two walls fall, the margin wall is frame-invariant (T112)

T112 (`adaptive_scaling_probe.py`, contract `ADAPTIVE.SCALING`, 20/20, SCALING-PARTIAL) stopped freezing the cell width and coupled it to the local scale. Two couplings were built: frame A, gap-coupled, $D_k = g_k/(2\nu)$ with g_k the local prime-power gap and ν the resolution constant of the frame; and frame B, mean-field, $D_k = \log n/(2\nu n)$ — the smooth PNT version of the same idea. The first finding is a structural asymmetry: *frame B has no ladder* (only 38% of consecutive pairs nest; the first failure is already at $n = 3$) — the nested handover lives exclusively in the gap coupling, not in any smooth $1/n$ law. T105 admissibility, re-measured in the scaled frame, sets a resolution floor $\nu \geq 4$ (at $\nu = 1$, 3/10 test zones are inadmissible). The flatness verdict is honest and double-edged: the ratio is *not* flat, but the exponent is *frame-invariant* — the components shift massively (m_k: $n^{-1.93} \to n^{-0.95}$; need109: $n^{-0.98} \to n^{+0.03}$, i.e. practically n-independent in the scaled frame), yet the difference stays -0.974.

The three walls, scaled. The **ladder wall is structurally gone**: with $D_k = g_k/(2\nu)$ the nested step has exactly ν cells per end *by construction* — twin pairs included — verified as an exact arithmetic lemma over all zones; the atom entry is the exact zero matrix by construction rather than by arithmetic accident; and the T111 killer pair $461 \to 463$ now certifies *spectrally* ($h = 1418$, retention 1.000000), with the reserve *opening*: $f_{\mathrm{crit}} = 8.1 \times 10^{-4}$–$4.2 \times 10^{-3}$ instead of 1.00 — roughly a factor 1000. (The pair $521 \to 523$ would need $h = 1634$, above the dimension cap: arithmetically co-certified, spectrally declared honestly as not run.) The ω **wall is no longer a depth wall**: the depth coefficient is $n^{-0.079\pm0.121}$, compatible with zero — ω_{cert} is driven by the resolution and the local gap, not by the depth of the ladder; a residual requirement-vacuum tail of 5 zones on the deepest ladder remains and is not booked as "gone". The **margin wall stays**, practically unmoved: the first sub-1 zone sits at $n = 449$ against the frozen frame's 461–463 (a factor 1.07) — it is not geometry but the substance of the T109 requirement itself.

The limit object. Overlaying the scaled windows, the archimedean part (0.657) and the pole part (0.558) contract Cauchy-like with amplitude laws $D^{-0.26}$ and $D^{+0.37}$; the atom part does *not* contract (0.935) — it is the prime-gap statistics itself, written in scaled coordinates. The spectral shape does not drift: $\lambda_2/\lambda_1 \sim n^{-0.029\pm0.052}$. The resulting formulation is a split object: *[a deterministic limit shape: arch + rank-one pole] + [the atoms as a controlled, non-convergent perturbation].* New and uncomfortable is the **regrid object**: the zone frames are not nested, so Rayleigh–Ritz transfers nothing between them; the regrid cost is measured at rate $(D'/D)^{+2.93}$ — an object that simply does not exist in the frozen frame, disclosed here because any limit argument has to pay it.

The trade — and the disclosed prime-gap dependence

Three arithmetic walls are exchanged for two analytic demands:

[P1] **Positivity of the limit operator** — *one* operator instead of a sequence of matrices; the deterministic part (arch + rank-one pole) contracts, the atom part is the perturbation to be controlled.

[P2] **A convergence rate below the step reserve** — the Cauchy rates and the regrid cost against the opening reserve f_{crit}.

The prime-gap inputs are disclosed rather than hidden. Upward, Bertrand–Chebyshev 1852 ($g_k \leq \log 2$; used and verified in-run) and the trivial even-gap bound suffice. The only remaining arithmetic parameter is the normalised gap $\theta_k = g_k\, n/\log n = 0.18$–$4.12$ — and downward, θ is *provably* unboundedly small: by Zhang 2014 / Maynard 2015 (infinitely many bounded gaps), $\theta \to 0$ on subsequences, so any gap-coupled construction must be uniform in $\theta \to 0$, at the price of the $1/\theta$ cost explosion.

T113 subsequently answered the currency question (next subsection): the wall is currency-invariant — and what it measures is the discretisation, not the spectrum.

7.12 The currency question: the wall is substance, and it measures the discretisation (T113)

T113 (`limit_operator_probe.py`, contract `LIMIT.OPERATOR`, 27/27, SUBSTANCE-CONFIRMED) answered the question T112 left open — is the margin fall in the scaled frame a normalisation artefact? — and the answer reinterprets the program's hardest number.

The currency table. The floor/requirement ratio carries the *same* exponent, $n^{-1.168\pm0.259}$, in all five currencies tested (raw, per $\lambda_{\max}$, per trace density, per D, per D^2) — and the invariance is structural, not accidental: the piecewise-constant basis is L^2-orthonormal (Gram matrix exactly I, so there is no free D-power to spend), and the zone matrices are restrictions of *one* quadratic form to subspaces of the same L^2 space (cross-grid form identity at relative 1.2×10^{-14}). The floor's homogeneity degree in the cell width is exactly 1.0; the requirement's is 1.0 under every legitimate rescaling. No flat currency exists. The wall is currency-invariant: substance, not bookkeeping.

Two surprises about what that substance is. First, the limit object is *not* "arch + pole with the atoms as a perturbation": the atom-free window form is deeply negative ($\lambda_{\min} = -3.08$ to -19.36, growing like $n^{+0.60}$), the atom block is equally large (norm 3.21–19.44), and the positive floor survives only as a *cancellation* of relative size 1.3×10^{-7}–9.7×10^{-5} — on the floor vector, -11.64 against $+11.64$. The Weyl bound is saturated: norm perturbation theory is five orders of magnitude too coarse for this object. Second, there is *no plateau under refinement*: at a fixed window, on nested grids over 8 levels (a factor 128 in resolution), $\lambda_1 \sim D^{1.83}$ and $\lambda_2 \sim D^{1.76}$ — the *same* power, with λ_2/λ_1 flat. **The continuum window form has no gap; m_k measures the gap of the discretisation.** This resolves the T112 tension exactly: λ_2/λ_1 constant while the floor falls is what it looks like when both eigenvalues carry the same D-power.

The atom part is almost invisible at the floor. The entering (deepest) atom contributes almost nothing to the coupling (8.3×10^{-11}–1.3×10^{-6}) — which is why handover steps certify at retention 1.000000 while the margin tears. The floor is almost pure geometry: $\log m =$

const $+ 1.87 \log D - 2.07 \log \alpha$, with an arithmetic residual of a factor 1.13 and no $\Lambda(n)$ dependence. The $\theta \to 0$ stress test separates cost from substance: the operator norm does *not* diverge ($\theta^{-0.005}$); an extra atom at the window edge shifts the floor by only 0.2–0.4%, while the same atom in the interior shifts it 10^7 times more — the $1/\theta$ explosion of T112 is purely a *cost* statement ($h \sim \nu n/\theta$ cells), not an operator instability.

Regrid, re-read. The raw regrid rate $(D'/D)^{3.85}$ is mostly the floor's own D-power — a normalisation jump, not an instability; the reserve certifies $f_{\mathrm{crit}} = 9.5 \times 10^{-2}$ (better than the T112 estimate), and after subtracting the D-power the reserve buys 15 steps (against 0 on the raw jump).

The new hardness balance — replacing [P1]/[P2]

The T112 trade is restated by measurement; this balance supersedes the earlier two-demand form.

[P1] **Semidefiniteness of the balanced form without a gap** — the continuum window form is semidefinite at best; strict limit-floor formulations demand what refinement demonstrably refuses.

[P2] **Certify the D-power itself** — a Grenander–Szegő-type consistency statement for $\lambda_{\min}$ of the scaled form; the power, not an abstract rate, is the object.

[P3] **New, and now leading: the requirement homogeneity** — need109 carries $D^{0.63}$ against the floor's $D^{2.38}$: the T109 chain divides by an *artifact* margin. The repair the measurements point at is a *margin-free step certificate* — a Schur/Loewner formulation certified from semidefiniteness alone, never dividing by a margin.

[P4] $\theta \to 0$ **is a cost-uniformity demand only** — the frame needs $h \sim \nu n/\theta$ cells, but the operator itself is θ-stable.

Disposition after T114 (next subsection): [P3] is **discharged** — the margin-free step certificate exists and runs past the wall; [P4] stands as a pure cost statement; a new **[P5]** appears (the (R) demand is grid-bound and is deleted, not transported); and [P1]/[P2] are restated as the two-object core — ε relative to κ, and Schur transport between grids.

T114 (`margin_free_step_probe.py`, contract `MARGIN.FREE.STEP`) then rebuilt the chain without the margin division — and the wall dissolved.

7.13 The wall dissolves: the margin-free step (T114)

T114 (`margin_free_step_probe.py`, contract `MARGIN.FREE.STEP`, 22/22, WALL-DISSOLVES) built the certificate that [P3] demanded, and the answer to the preregistered question — does the margin wall disappear as a requirement artefact? — is yes.

The margin-free step. The structural fact that makes it possible was already on the table (T110/T111): the restriction of the grown window's form to the old window is bit-exactly Q_{old} — the entering atom's block is the exact zero matrix — so margin-freeness *is* that property, once the step stops asking for more. Route (i) is Albert 1969: the generalised Schur criterion with the Moore–Penrose inverse,

$$Q' = \begin{pmatrix} A & C \\ C^{\mathsf{T}} & X \end{pmatrix} \succeq 0 \iff X \succeq 0, \quad \mathrm{ran}(C^{\mathsf{T}}) \subseteq \mathrm{ran}(X), \quad A - CX^{+}C^{\mathsf{T}} \succeq 0$$

(the range condition is Douglas 1966) — no margin appears anywhere. It certifies 27/27 ladder steps, *eleven of them beyond the old wall* (467, 479, 512, 529, 547, 577, 661, 773, 887, 1129, 1331; deepest $h = 1495$). The exact 4×4 Schur complement is an $O(0.1)$ object with *no* cancellation: $\lambda_{\min}(S) = 0.068$–$0.154$, i.e. 42–67% of the block scale. Route (ii), the T110 graded minorant in its degenerate limit $x_{\mathrm{in}} = 0$, dies structurally on 27/27 — which retypes T110's margin: it was not paying for positivity but for an *abandoned direction.* Route (iii), unshifted Cholesky, passes 27/27 against the declared backward-error floor. Negative controls are sharp (5/5), and the certificate is frame-invariant at $\nu = 8$ (3/3). All seven zones at which the T109 chain tears — including the wall zone $n = 449$ itself, where need109$/m = 1.241$ — are certified by the margin-free step.

The wall in one line. The same quantity, bounded through norms (Weyl: $\lambda_{\min}(A) - \|C\|^2/\lambda_{\min}(X)$), is *negative* by a factor of 2.4×10^5–9.6×10^7: an $O(1)$ numerator divided by the 10^{-6} artifact floor. Every norm route had to fail — and the exact Schur complement passes through none of it.

The favourable power direction. On nested refinement chains at a fixed window, $\varepsilon \sim D^{1.77\text{–}2.01}$ and $\kappa \sim D^{3.64\text{–}4.84}$ (fits) — the preregistered "same power" expectation is refuted *in the favourable direction*: κ falls faster, so the scale-free ratio $r = \kappa/\varepsilon \sim D^{1.6\text{–}3.1}$ shrinks under refinement. The ratio closure $r = 2 \times 10^{-4}$–0.103 holds on 27/27 steps (11/11 beyond 463). The circle [base semidefiniteness (a declared hypothesis input) + margin-free step + ratio closure] closes 27/27; ten multi-step chains run (21 certified steps, the longest of length 4); and *every* chain ends at the cap $h \leq 1500$ — a window *cost* — never at a step. Regrid honesty: 0/464 gap ratios are dyadic, so the non-nested transport object of T112/T113 remains real.

[P5]: the (R) demand is grid-bound. Holding the physical demand $(\mu/2)P_-$ fixed and refining the grid, ω grows monotonically (5/5) and crosses 1 (4/5): the (R) formulation is *false in the continuum* — it is to be *deleted*, not transported. The exact Schur complement needs no demand surrogate in its place.

Cancellation robustness. The 10^{-7}-relative cancellation of T113 has 5.3–8.8 decimal digits of headroom above the Cholesky backward-error floor (Wilkinson/Higham), and it sits almost entirely in the rank-one pole — whose positivity is exactly the T108 Szegő identity ε.

The new core — two objects

The four-part balance collapses. [P3] is discharged (the margin-free step certificate exists and runs past the wall), [P4] was only ever a cost statement, and [P5] deletes (R). What is left:

[P1] $\varepsilon > 0$ **with controlled size relative to** κ — the exact Szegő/Levinson identity of T108, now the single positivity object of the program.

[P2] **Transport of the exact Schur complement between non-nested grids** — an $O(0.1)$-sized object with no cancellation, sharpened from the earlier abstract D-power demand.

Disposition after T115 (next subsection): [P2] **splits** — the transport across a real regrid certifies only mild refinement ($\rho \lesssim 1.8$), for a principled reason (the Schur floor itself falls under refinement), while a two-scale compression breaks the compute cap; [P1] **sharpens** to one scalar inequality, the classical Szegő–Levinson prediction-error bound. The core is restated as the three-point list at the end of the next subsection — only one point of which is an inequality.

T115 (`schur_transport_probe.py`, contract `SCHUR.TRANSPORT`) then ran the two last bricks — and split them.

7.14 Transport blocked, cap broken: the two-scale compression (T115)

T115 (`schur_transport_probe.py`, contract `SCHUR.TRANSPORT`, 26/26, TRANSPORT-BLOCKED — with a large compression gain) took the two-object core apart: transport of the exact Schur complement across the ladder's non-nested grids, and a multi-resolution compression of the already-certified interior.

The transport splits at $\rho^* \approx 1.8$ — for a principled reason. Haynsworth's partial minimisation $y^{\mathsf T} S y = \min_z [y; z]^{\mathsf T} Q' [y; z]$ makes $\lambda_{\min}(S)$ available without any inverse and without a margin; the two-sided transport bracket built from it is an *identity* (bookkeeping), with the evidence sitting in the consistency defect η and in the lower end. On the eleven real regrid pairs of the ladder ($\rho = 1.023$–3.425) the lower end is positive on 7/11, with a clean split: every pair with $\rho \leq 2.061$ certifies and none with $\rho \geq 2.291$ does; a synthetic scan puts the threshold at $\rho^* = 1.83$, against a median real refinement ratio of 2.001 — 39% of the ladder's 7686 refinement pairs are covered. The blocker is *not* the transport machinery: the exact cell-overlap projection beats three deliberately wrong transfer maps on 9/9 (by factors up to 270), and on *nested* ladders — where the transport error is exactly zero — $\lambda_{\min}(S)$ itself falls like $\rho^{-1.73}$ to $\rho^{-1.65}$. The drop is real, so no bound can undo it. Two sharpenings came free: non-nestability is a matter of *integrality*, not dyadicity (0/14 807 gap ratios are integer, the closest miss 3.4×10^{-5}), and the measured defect $\eta = 2.7 \times 10^{-2}$–$8.1 \times 10^{-2}$ is 10^3–10^6 times smaller than its a-priori Céa/Strang norm surrogate —

which is exactly where the margin division would have re-entered.

The compression breaks the cap. The two-scale window (interior merged into blocks, the boundary strip kept fine, the merge anchored at the centre) preserves $X_{\text{mixed}} = Q_{\text{old,mixed}}$ *exactly* (rel 0.0) — the compressed step is still margin-free — and Albert certifies 66/66 (zone, q) combinations up to $q = 64$, with the compression error certified *one-sided and second order* in the projection defect (Rayleigh–Ritz plus stationarity of the fine minimiser; Céa/Strang; 66/66) and savings down to $m/h = 0.043$. The deep run then breaks the T114 cap: the margin-free step certifies at $n = 155\,921$ (117× T114's 1331), on a fine lattice of $h = 93\,470$ cells (62× the hard cap $h \le 1500$) compressed to $m = 1490$; re-coarsening inside a chain is free (Rayleigh–Ritz, rel $\le 2.1 \times 10^{-12}$); the longest chain is now *ten* steps (T114: four), with 33 certified steps over four chains; the stopper is the cost cap on 3/4 runs and a failing step on 0; and every certified quantity sits 10^5–10^{11} times above the declared Cholesky backward-error floor. Stated honestly: $\lambda_{\min}(S_{\text{mixed}}) \approx 5.1$ is 52× the uniform value because the coarser Galerkin space sees fewer soft directions — a weaker statement about the continuum, declared as such, and flat across the whole deep band. The contiguous reach an induction needs moves from $n \le 125$ to $n \le 5437$ (factor 43.5): the $n \log n$ cost barrier is *divided*, not broken.

The remaining list — three points, one inequality

The shortest list the program has produced. From T114's balance, the margin (dissolved), the (R) demand (grid-bound, deleted), the floor/ε transport (the object to transport is $\lambda_{\min}(S)$, the slowest-falling of the three) and the compression risk (it does not break the step) are all *struck*. What is left:

(1) **[P1] as one scalar inequality** — given $T = Q + \tilde{t}\tilde{t}^{\mathsf{T}} \succ 0$, positivity of $Q|_{\text{odd}}$ is *equivalent* to $\varepsilon = 1 - \tilde{t}^{\mathsf{T}} T^{-1} \tilde{t} \ge c(D) > 0$: the classical Szegő–Levinson prediction-error bound for one explicit symbol (the Haynsworth double-Albert device: $T \succ 0 \wedge \varepsilon \ge 0 \iff Q \succeq 0$). This is the target-sentence candidate. Measured: $\varepsilon \sim \rho^{-1.71}$, falling slightly faster than $\lambda_{\min}(S) \sim \rho^{-1.69}$ (fits).

(2) **An a-priori bound on the transport defect** η — a regularity statement about the minimiser z^*; it would turn the measured transport into a certificate for $\rho \le 1.8$.

(3) **A boundary formulation** (suggested by the exact-zero lemma) — never represent the old interior at all; it would remove the compute cap *and* the regrid at once.

Only (1) is an inequality; (2) is a regularity statement, (3) is a reformulation.

Disposition after T116 (next subsection): (3) is **run and splits** — the boundary recursion itself is exact (the pole rides in the state with no truncation, and the march cost is flat), but the state cannot be *bounded*: the prime comb refuses compression. (2) is **devalued** — a single boundary march needs no regrid, so the transport defect no longer sits on the critical path. (1) **sharpens** into a two-variable law with a named classical proof technique: ε is a Galerkin error, Aubin–Nitsche duality predicts the measured D-exponent, and the jumps at prime-power entries become an explicit side-condition.

T116 (`boundary_formulation_probe.py`, contract `BOUNDARY.FORMULATION`) then ran the reformulation — and it held exactly where the contract feared it would break, and broke exactly where it seemed free.

7.15 The boundary recursion: exact pole, flat cost, and the prime comb (T116)

T116 (`boundary_formulation_probe.py`, contract `BOUNDARY.FORMULATION`, 33/33, RICCATI-PARTIAL) executed point (3) of the list: never represent the old interior at all — run the induction as a pure boundary process.

The recursion stands, and the pole is free. The state is the triple (Σ, r, σ) obtained by partially minimising the pole-free form over the interior (Haynsworth 1968): the identity $t^{\mathsf{T}} T^{-1} t = r^{\mathsf{T}} \Sigma^{-1} r + \sigma$ holds on six zones at rel 6.7×10^{-10}, the Haynsworth double-Albert equivalence certifies 6/6 with sharp negative controls, and the Riccati-type chaining of two eliminations is *exact* (synthetic 1.4×10^{-16}; bit-exact on real windows). The first feared obstruction — pole bookkeeping — dissolves for free: the rank-one pole is genuinely *global* (its boundary value at half depth is 38.6%, against 0.2% for the archimedean part), but the pair (r, σ) carries it *exactly* through bordered elimination, with no truncation anywhere.

The tail bookkeeping breaks — on the prime comb. The second feared obstruction was the archimedean kernel tail, and it is innocent: the arch part decays cleanly, $\exp(-1.73\,\omega)$. What does not decay is the *full* symbol — it stays at $\geq 93\%$ of its maximum beyond every tested width, and 100% of the off-band mass sits in the stripes of the two *prime combs*, carried by the reflection comb: every incoming cell couples through a Hankel term to the interior cell at $\alpha - \log n_j$, for *every* prime power $\sqrt{n} < n_j < n$. A truncation of the state would have to discard 1.3×10^2–1.1×10^6 times the margin ε it must respect, a budget that grows like $n^{2.17}$ ($\mu_j \sim n^{-1/4}$ against $\varepsilon \sim n^{-1.8}$; fits); 11/18 real truncations break outright; a comb-aware state that keeps the stripes still spans 83.6% of the window and reaches only rel 20; and the ceiling of the whole idea is a $\Theta(\log^2 n)$ gain. There is no sparse faithful state — the stripes drift with fill-in.

The deep boundary run — a cost-geometry demonstration, declared as such. One unbroken chain of 169 236 prepends ran to a window of $h = 1\,354\,088$ cells — 903× the old cap — with the entire state $12 \times 12 + 12 + 1$ numbers and the cost *flat*: 76 μs per step, first-to-last-decile ratio 1.00. Stated honestly, this is *not* a Weil certificate: at the fixed cell width $D = 1.9 \times 10^{-6}$ the band contains no atom, the stopper is $\varepsilon < 0$ at $h = 1.35 \times 10^6$ — the *band model* loses positivity there, not the machine — and the atom reach stays at $n = 173$. What the march certifies is the cost geometry of the boundary formulation.

The one inequality, sharpened into a law with a proof technique. The measured law is $\varepsilon \approx 8.34\,D^{1.790}\alpha^{-6.04}$ (jackknife errors ≤ 0.041, rms ≤ 0.097; φ from a fixed-D sweep, ± 0.08; all fits). ε is *not* smooth in α: it jumps by factors up to 120 at prime-power entries, so any smooth ansatz is wrong from the start. (T117 subsequently resolved the same sweep cell by cell and *corrected* this reading: the entries themselves move ε *up*, and the factor-120 falls are the accumulated smooth bordering decay between sweep points — see the T117 subsection. The criticality and D-law statements of this paragraph are unaffected.) The continuum form is *exactly critical* — $\tilde{t}^{\mathsf{T}} T^{-1} \tilde{t}$ reaches 0.999975745, and Szegő–Kolmogorov says the pole is representable in the limit — with every measurement $\geq 1.1 \times 10^5$ above the declared Cholesky floor. The classical hit: in the Galerkin reading, $\tilde{t}$ represents the D-independent function $2\sinh(x/2)$, ε is the error of a piecewise-constant Galerkin method for it, and Aubin–Nitsche duality predicts $\varepsilon \sim D^{2s}$: the measured $\theta = 1.79$ gives $s \approx 0.90$, exactly the regularity a log singularity plus window edges would produce. The target inequality is now formulated: $\varepsilon(\alpha, D) \geq c_0\, D^\theta \alpha^\varphi$, uniformly across the jumps.

The remaining list after T116 — one inequality, one obstruction

Won: the exact pole (no truncation), the exact Riccati chaining, and flat cost per step. Broken: [**SUPPORT**] — the prime comb; there is no sparse faithful state, and the stripes drift with fill-in. Devalued: [**P2**] — a single boundary march needs no regrid, so the transport defect leaves the critical path. The basis is uncritical. What remains on the critical path:

(1) **[P1] as a two-variable law** — $\varepsilon(\alpha, D) \geq c_0\, D^\theta \alpha^\varphi$ with $\theta = 1.790$ and $\varphi = -6.04$ (fits), to be proved *across* the factor-120 jumps at prime-power entries; the proof technique is named (piecewise-constant Galerkin error for $2\sinh(x/2)$, Aubin–Nitsche duality, $s \approx 0.90$).

(2) **The comb as the honest support obstruction** — any bounded-state formulation must either carry the reflection stripes ($\Theta(\log^2 n)$ ceiling) or find a representation in which the comb is diagonal.

Disposition after T117 (next subsection): (1) is **theorem-shaped** — the Galerkin reading becomes four exact identities, the lower-bound direction acquires a certified two-level chain that loses no power of D, and the jumps acquire exact closed forms with the *opposite* sign: entries *raise* ε, the factor-120 falls were a sweep artefact, and the jump side-condition is *removed* from the ansatz. What is left of (1) is three named analytic lemmas about one symbol, two of them constants. (2) is untouched.

T117 (`epsilon_theorem_probe.py`, contract `EPSILON.THEOREM`) then took the inequality apart — keeping the one honest asymmetry visible everywhere: the classical Galerkin machinery produces *upper* bounds for approximation errors, and the target needs a *lower* bound.

7.16 The one inequality becomes theorem-shaped (T117)

T117 (`epsilon_theorem_probe.py`, contract `EPSILON.THEOREM`, 23/23, THEOREM-SHAPED) converted the T116 reading into identities and a certified chain. The direction problem is stated once and kept visible: Céa 1964, Aubin 1967/Nitsche 1968 and Bramble–Hilbert 1970 all bound a Galerkin error from *above* by an interpolation error; the missing lower-bound ingredient carries a classical name — the *saturation assumption* (Bank–Smith 1993; Dörfler–Nochetto 2002) — and is known to be an assumption, not a theorem. Every classical anchor below is therefore cited with the direction it actually gives.

The Galerkin identity, exact. Four identities, verified on six windows to 10^{-10} relative or better, none of them asymptotic: (i) $\tilde{t}$ is the cell functional of the *D-independent* function $f(x) = 2\sinh(x/2)$ (independent quadrature, rel 5.2×10^{-16}); (ii) the family is *exactly nested* — with P the block-averaging prolongation, $T_c = P^{\mathsf{T}} T_f P$ and $\tilde{t}_c = P^{\mathsf{T}} \tilde{t}_f$ (rel 2×10^{-14}) — so ε is a Galerkin best-approximation error of *one* bilinear form on a nested family, and its monotonicity under refinement is a *theorem*, not a fit; (iii) $\varepsilon = \min_v [\, 1 - 2\tilde{t}^{\mathsf{T}} v + v^{\mathsf{T}} T v\,]$ with the two-level Pythagoras $\varepsilon_c - \varepsilon_f = \|u_f - P u_c\|^2_{T_f}$ (rel 6.2×10^{-11}); (iv) the dual-norm residual form $\varepsilon_c - \varepsilon_f = \|\tilde{t}_f - T_f P u_c\|^2_{T_f^{-1}}$, whence $\varepsilon > 0 \iff \tilde{t} \notin T(V_c)$: positivity is a *non-membership* statement — which is exactly why the quantitative version is hard, since a non-membership statement carries no rate by itself.

The lower bound, honestly. The crude psd-minorant route ($\varepsilon \geq 1 - \tilde{t}^{\mathsf{T}} G^{-1} \tilde{t}$ for $0 < G \leq T$) is measured dead: criticality forces any useful minorant to be sharp to relative 8.3×10^{-4} in the one direction $u = T^{-1}\tilde{t}$ — which is the same as knowing u. What survives is the *two-level chain*

$$\varepsilon_c \;\geq\; \varepsilon_c - \varepsilon_f \;=\; y^{\mathsf{T}} S y \;\geq\; \lambda_{\min}(S)\, \|y\|^2 \;\geq\; \lambda_{\min}(S)\, D_f^3 \max_j g_j^2/2 \;>\; 0,$$

with y the oscillation of the fine dual solution, S the T-Schur complement onto the oscillation space (Haynsworth 1968), and the last step the one-cell Payne–Weinberger 1960 cell-variance inequality used in its *lower* form — the only place where a classical approximation statement points the right way. Every link is an identity or a completed Cholesky: certified on all 19 nested pairs, bound/$\varepsilon \in [0.111, 0.185]$, everything at least 6.7×10^5 above the declared backward-error floor. The rate: $\theta'(\text{sum}) = 1.74$ against $\theta = 1.76$ (jackknifed fits) — a loss of 0.04 powers of D, i.e. *none*, because $\lambda_{\min}(S)$ *grows* under refinement (+0.20 per halving of D; log-versus-small-power undecidable on an $8\times$ lever, and nothing in the chain rests on deciding it): the chain costs a constant, not a power (bound/ε spread 1.195 over the whole lever). The one-cell version costs a further 0.93 powers — the classical price of replacing a sum over cells by its largest term.

The jumps, exactly — and a correction to T116. Growing the window by one cell per end at fixed D is a pure prepend in frame A, so the bordered inverse gives the *exact* drop $\varepsilon(h{+}1) = \varepsilon(h) - r_0^2/s_0$ (Levinson 1947; rel 3.7×10^{-12} over consecutive steps). A prime-power entry at fixed window is a *corner* update supported on the two outermost Hankel anti-diagonals, of rank exactly $k_0 + 1$ (k_0 the corner distance), and Woodbury/Sherman–Morrison reproduces its effect on ε in closed form on all 23 measured entries (rel 2.1×10^{-11}). The direction *corrects* T116: all 23 entries move ε *up* (factors 1.0003–1.0506; for $k_0 = 0$ the update is $+\beta e_0 e_0^{\mathsf{T}}$ with $\beta > 0$, so an entry can never by itself lower ε at that corner distance). The T116 factor-120 falls were a *sweep artefact*: resolved cell by cell over 400 steps, the largest single-cell drop is a factor 0.9555, the entry share of the total log-drop is −0.3% *with the opposite sign*, and the falls are the accumulated smooth Levinson product between sweep points spaced 150–300 cells apart. The jump side-condition disappears from the ansatz; what a bound must survive is a smooth $\alpha^{-6.07\pm0.03}$ decay (fit; the α lever is short and bound by the compute cap $h \leq 1400$, declared as such).

The theorem candidate — and the three-lemma missing list

Candidate. Let $V_c \subset V_f$ be the reflection-odd piecewise-constant spaces on $(-\alpha, \alpha)$ with M and $2M$ cells, T the pole-free odd Weil form, $\tilde{t}$ the pole functional of $f(x) = 2\sinh(x/2)$. Assume (H1) $T_f \succ 0$ [certified per window]; (H2) $\lambda_{\min}(S) \geq \kappa > 0$ on the oscillation space [certified per window; measured to grow under refinement]; (H3) one coarse cell carries a non-vanishing discrete slope of the dual solution [measured]. Then the four-line chain above gives $\varepsilon_c \geq \kappa\, D_f^3 G^2/2 > 0$; at fixed D the α-decrease per added cell pair is exactly r_0^2/s_0; and prime-power entries only raise ε — so the certified bound carries the full measured rate. Line-by-line attribution: the identities are exact algebra, (H1)/(H2) are completed Choleskys, the cell step is classical, the exponents are jackknifed fits.

What is still missing — three named analytic lemmas about one symbol, each with a classical address:

(1) a uniform lower bound on *one* local slope of the dual solution $u_f = T_f^{-1}\tilde{t}_f$ (H3 as analysis): the corner asymptotics of T^{-1} for a symbol with a log singularity — the Kac–Murdock–Szegő 1953 / Widom 1974 circle of results;

(2) a lower bound for $\lambda_{\min}(S)$ at *one* level (H2 as analysis): S is the form seen by oscillations, i.e. the symbol at the Nyquist frequency π/D, where the archimedean weight is largest — the measured monotonicity then carries it;

(3) the saturation constant: dropping ε_f costs the factor $1 - \varepsilon_f/\varepsilon_c$, measured in $[0.675, 0.744]$ — literally the Bank–Smith / Dörfler–Nochetto saturation assumption, bounded away from 0 by a constant, so it costs no rate.

Two of the three are *constants*, not rates. None of this touches the T116 [**SUPPORT**] comb obstruction — that is a separate question about *state compression*, not about ε.

Status after T118 (next subsection): (3) is **classically done** — saturation is an *identity* here, $\varepsilon_c - \varepsilon_f = y^\mathsf{T} S y$, plus the one hypothesis Bank–Smith themselves assume; (2) is **certified on windows and arithmetically reduced** — not through the symbol of S itself, which is a weighted *harmonic* mean of the aliasing pair and vacuous on the comb, but through the strengthened Cauchy–Schwarz shift onto the oscillation Gram; (1) is **corrected and open** — the corner exponent does not settle, because a log singularity is the boundary of all powers, and (H3) must be upgraded to a mean-square statement, which also repairs the chain's last step.

T118 (`symbol_lemmas_probe.py`, contract `SYMBOL.LEMMAS`) then measured the three lemmas against their classical addresses, one by one — and two of the three stand.

7.17 Two of the three lemmas stand (T118)

T118 (`symbol_lemmas_probe.py`, contract `SYMBOL.LEMMAS`, 36/36, TWO-OF-THREE) put each missing lemma against its classical address. The outcome, stated before the detail: the saturation lemma (3) is an *identity* here; the Schur-floor lemma (2) acquires a fully certified classical route on half the measured windows and an arithmetic reduction on all of them; the corner lemma (1) does *not* contradict its address — the address simply does not cover this symbol — and its failure *corrects* both an old exponent estimate and the last step of the T117 chain. What the theorem still needs is exactly one genuinely new analytic statement.

The harmonic versus the arithmetic aliasing mean. The two-level oscillation Schur complement of a Toeplitz form has an *exact* two-grid symbol (re-derived in closed form; local Fourier analysis, Brandt 1977/Hackbusch 1985; multigrid for Toeplitz, Fiorentino–Serra 1991), and it is the weighted *harmonic* mean of the symbol at a low frequency and at its Nyquist partner: $1/\sigma_S = \cos^2(\theta/2)/f(\theta+\pi) + \sin^2(\theta/2)/f(\theta)$. A harmonic mean needs *both* entries positive, and f is negative on a comb of dips at the window's own resolution scale: $\inf \sigma_S = 0$ on 14/14 (zone, level) rows — every pointwise route through σ_S is *vacuous* on the real symbol, not merely weak (a perturbative split recovers at most 2.6×10^{-5} of $\lambda_{\min}(S)$; the comb is non-perturbative). The classical reason is computed rather than asserted: $\lambda_{\min}$ at finite M is a Fejér–Riesz *moment* problem, not a pointwise infimum — the ground state of S *concentrates* on the negative set (3.3–43× the uniform share), and positivity is a cancellation of relative size 10^2–10^4 between the positive and negative parts of the symbol integral. Coarse-graining shows where the mass sits: the one-cell-averaged symbol is positive on 14/14 windows (a heuristic, declared as such). The growth is a *logarithm*, not a power: on an FFT-only lever of up to 128× in D the log model beats the power model (at $b = 0.44$–0.93 per $\log(1/D)$), exactly what the log-singular kernel demands (Kac–Murdock–Szegő 1953; Widom 1974). And the pole mass in the oscillation space is $\leq 2.1 \times 10^{-5}$ — an identity via the DTFT factor $|\sin(\theta/2)|$ — which is why the finite-section effect is absent there.

Saturation is an identity here — and the CBS shift rescues Lemma B. For a nested piecewise-constant pair, saturation is not an assumption but the *identity* $\varepsilon_c - \varepsilon_f = y^\mathsf{T} S y$ (rel 4.9×10^{-11}), so β is *computed*: $\beta \in [0.252, 0.336]$ over 20 triples, with the monotonicity in the favourable direction; what remains of Lemma C is the one hypothesis Bank–Smith 1993 themselves assume. Their strengthened Cauchy–Schwarz constant γ (Yserentant 1986) is then the lever that rescues Lemma B: the identity $\lambda_{\min}(L_z^{-1} S L_z^{-\mathsf{T}}) = 1 - \gamma^2$ (rel 2.1×10^{-15}) moves the symbol question off S and onto the oscillation Gram, whose symbol $\sigma_z(\phi) = \sin^2(\theta/2) f(\theta) + \cos^2(\theta/2) f(\theta + \pi)$ is the *arithmetic* mean of the same aliasing pair — the low-frequency negativity that poisoned the harmonic mean is suppressed *quadratically* by $\sin^2(\theta/2)$. The route is then certified: $\inf \sigma_z > 0$ on 7/14 windows (systematically: every window with $\alpha \le 1.30$ and $h \ge 200$), with the certified bound recovering up to 52% of the measured $\lambda_{\min}(S)$; on *all* 14 windows the reduction $\lambda_{\min}(S) \ge (1 - \gamma^2)\, \lambda_{\min}(T_h(\sigma_z))$ holds at 32–76% sharpness — Lemma B is reduced to the positivity of a plain Toeplitz section of an explicit symbol; and on the long FFT lever (M up to 15 680, no matrix ever formed) the certified floor $\inf \sigma_z$ *rises* logarithmically and *crosses zero* on 3/5 zones — the failing windows were *under-resolved*, not obstructed.

The corner lemma corrects itself — and repairs the chain. The classical statement is an edge exponent: for a symbol vanishing like $|\theta|^{2s}$ the finite-section solution vanishes like $\operatorname{dist}(\cdot, \text{edge})^s$. The estimator is calibrated on a model class where this is exact ($|2\sin(\theta/2)|^{2s}$, lags in closed form; the Kac–Murdock–Szegő 1953 closed-form inverse as the $s = 0$ control). On the *real* symbol the corner exponent *drifts*: $+0.238$ per halving of D (3–7.4σ) — $\log(1/|\theta|)$ is the boundary of all powers, so there is no asymptotic (s, κ) to cite. (The old T108/T109 24-cell exponent was doubly mis-estimated; its apparent stability was the cancellation of two errors.) The (H3) correction: the single-cell form is mis-dimensioned — the concentration factor grows like $h^{0.99}$, the slope mass sits on a fixed *fraction* of cells — and the repair is at the T117 chain's last step: the rate lives on $\|y\|^2$ (the stage-two bound carries $D^{1.751}$ against $\varepsilon \sim D^{1.770}$ — lossless; the single-cell stage costs $D^{2.74}$), so (H3) must be a mean-square statement.

The theorem after T118 — five unconditional links, two hypotheses

Candidate (V2). Five unconditional links — the saturation identity $\varepsilon_c - \varepsilon_f = y^\mathsf{T} S y$, the CBS identity $\lambda_{\min}(L_z^{-1} S L_z^{-\mathsf{T}}) = 1 - \gamma^2$, the isometry ordering onto $T_h(\sigma_z)$, Grenander–Szegő in the lower direction, and the certified Cholesky floors — give, for every window the probe touched,

$$\varepsilon(\alpha, D) \;\ge\; (1 - \gamma^2)\, \lambda_{\min}\big(T_h(\sigma_z)\big)\, \|y\|^2.$$

Two hypotheses close it: **(H1)** γ bounded away from 1 uniformly in (α, D) — the standard CBS hypothesis of the two-level literature, measured $1 - \gamma^2 \ge 0.181$ with the favourable trend; **(H2)** $\|y\|^2 \ge c\, D^{1.75}$ uniformly — (H3) upgraded to mean square, and *the* one genuinely new analytic statement the theorem needs, not importable from the Kac–Murdock–Szegő/Widom power-law corner theory because this symbol is log-singular with atoms.

The per-lemma missing lists. **L-B** (two points, both *arithmetic*): $\inf \sigma_z > 0$ for all $D < D_0(\alpha)$ — D_0 set by the comb depth against the Nyquist-band mass, an atom-counting question — and narrow-dip positivity of finite sections. **L-C** (one point): γ-uniformity, i.e. (H1). **L-A** (two points): the mean-square form (H3′), and the Fisher–Hartwig extension of the corner theory to log-singular symbols with atoms.

T119 (`oscillation_mass_probe.py`, contract `OSCILLATION.MASS` — the arithmetic half of L-B (the $D_0(\alpha)$ formula, deep verification), the (H2) section, the (H3′) mean-square form, and a theorem candidate V3) was running at the time of writing and is *not* reported here.

8 The kill list

A program that only reports what survived is not reporting. These are the readings and routes that were refuted *by the program's own probes*, each with the measurement that killed it. They are listed as a section rather than as footnotes because they are the more transferable output: each one tells a later attempt where not to go.

K1. The essential-singularity reading of the onset (T96, refuted T102). T96 read the counterfactual onset as super-polynomial, compatible with $\exp(-c/\delta')$, i.e. an essential singularity. T102 showed the onset is *anchored* at the crossing $\delta_c = 2w_k$ of two finite quantities ($R^2 = 0.968$), that exponential versus power behaviour is not separable at this resolution, and that the scatter of c (CV 103%) collapses to $c' = 1.52$ in the scale-free variable. The essential-singularity reading is *compatible but not distinguished* — and it is no longer used.

K2. The C/g law as a causal law (T101, refuted T102). Triple negative: g_k is causally impossible (Q_{k-1} is blind to atom $k+1$), statistically dispensable (causal law $C \cdot u$: CV 23.3% versus 27.6%), and arithmetically only a ceiling ($C_k \le g_k \mu_k$ exactly; the 16-zone extrapolation violates its own ceiling from $k = 69$; checked on 18 120 prime-power atoms to $n = 200\,000$). *It was a proxy.*

K3. The naive margin route (T104). $M \ge m \cdot \mathrm{Id}$ holds in 0/16 zones; the coupling-mass to margin ratio is 10^3–10^6; $m \sim M^{-1.7}$ vanishes in the continuum (fit). The genuine form is nearly singular on the induction window and no margin rescues it. (The prolate wing basis recommended by T96/T103 was tested in both arms and also rejected.)

K4. The density chain (T106). It would need a hard gap $w_0 \ge 0.55$–2.44; measured 5.3×10^{-6}–1.1×10^{-3} — short by a factor 5×10^2 to 3.6×10^5. Dead for good.

K5. The Landau–Widom anchor (T106). The wrong classical anchor for this problem (rms 0.50–1.55). The right one is Grenander–Szegő on the Planck-coarsened symbol — averaging over the mode-spacing cell π/M — which hits 0.92–2.43 and is better on 67/72 pairs. This also explains why a negative symbol (dip $-21.3 \ldots -3.4$) still gives $Q \succeq 0$: the atoms oscillate exactly at the window's Planck scale.

K6. The accumulated invariant with self-propagation (T106). A no-go *with a measured mechanism.* The wing demand gives $\beta_{\text{wing}} = 2.21$–$14.43 > 1$ on 16/16, but inherited demands give $\beta_{\text{int}} \sim 10^{-2}$ on 15/15: an accumulated invariant with absolute anchoring is false. The reason is transport — in a larger window an old wing pair becomes an interior pair and draws 99.7–99.95% of its demand from the 64 softest modes (coupling 751–2653× stronger). No self-propagation either ($\beta_{\text{on}} < 1$ on 15/15, loss 13–320×, $\theta_k = 0.9866$–0.9998). *The wing margin is an edge property of the window, not a property of the pair.* What survives is the window-wise family $\sigma_k(\delta_0) \ge 1.31(\mu_k/2)$, uniform in M with drift $\le 2.01\times$.

K7. The symbol route for the odd channel (T107). Structurally dead, not merely weak: the symbol lower bound for T_{odd} is certified but *negative* ($\lambda_{\text{cert}} = -2.54 \ldots -0.81$) against a measured $\lambda_{\min}(T_{\text{odd}}) = 4.7 \times 10^{-5} \ldots 6.4 \times 10^{-2}$ and a symbol infimum of $-22 \ldots -6$: five orders of magnitude and a sign change. The soft edge of T_{odd} is a pure finite-section effect that no symbol can see. Coarse certified chains (Cauchy–Schwarz against $\lambda_{\min}$) close 0/16 — and fail even with the *measured* $\lambda_{\min}$, so it is a category error, not a precision problem.

K8. Three certificate candidates for the avoidance norm (T108). Killed *quantitatively*, which is the useful form: the cancellation route ($\sum |k| = 401$ against $\sum k = 1$), the Bessel route (off by 10^7) and the pole Cauchy–Schwarz route (off by 10^6). ω_{cert} is blocked by the compression Schur distance alone (0.18–0.52). What replaced them is not a fourth bound of the same kind but a different object class: a boundary-decay statement about one explicit vector. T109 then killed the natural fourth candidate as well: **the Combes–Thomas resolvent bound at the boundary**, evaluated with the *exact* Green row so that no constant can rescue it, closes 1/16 — the mechanism at the edge is cancellation (44–4.6×10^4 on 15/16 zones), which no absolute-value bound survives. What works instead is the residual certificate that carries the cancellation (Section 7).

K9. The scalar step law for the margin (T110). Killed structurally, in three pieces: bordered Weyl closes 0/15 handovers; the Schur/Friedrichs scalar cap is vacuous on 15/15 (its scalar-only bound degrades back to Weyl); and the strictly scalar graded minorant ($n_{\text{soft}} = \dim$) closes

0/15, with the strictly scalar *induction* compounding to a Schur factor of 2.7×10^{-36} by $k = 3$. The reason is a boundary layer: the new cells of a grown window attach at the edge where the pole source sits, so a one-number summary of the bordering is blind to the deciding direction. What survives is the graded ($n_{\text{soft}} = 1$) Loewner minorant — one soft direction is exactly the amount of geometry the step needs.

K10. **Two earlier structural kills**, recorded here because they still bound the design space: the cross-block $\sqrt{\lambda_0}$ law is Douglas range inclusion and therefore circular (T98), and the cone-library route saturates at 5/24 coverage and is dead as a covering strategy (T72, promoted into v540 as a named limit). Likewise the seam route to the archimedean place (T20–T25) and the Kohnen-plus route to the central values (T44, promoted into v537 as a scope fence).

9 Status and outlook

9.1 What is certified, what is measured, what is hypothesis input

The three-way split, stated once

Certified (exact identities or closed-form bounds, machine-checked): the seven ledger modules of Section 4; and, in the sandbox arc, the bare bound b_0 in closed form, the parity superselection $JQJ = Q$, $JB = -BR$, the pole splitting $ab^\top + ba^\top = ss^\top - tt^\top$, the channel inertia, the avoidance law $Q_{\text{full}} \succeq 0 \Rightarrow \|B^\top v_i\|^2 \leq w_i \Lambda_-$, the growth inequality, the accumulation bounds, the Sherman–Morrison/Woodbury reduction of (R) to $r \leq 1$, the three T108 identities (ε as a $Q|_{\text{odd}}$ energy, the rank-one overlap $hh^\top/\varepsilon$, and ε as the square of the last Cholesky pivot), the two T109 certificates (the graded matrix cap $\omega_{\text{cert}} = 0.2561$–$0.7569$, no hypothesis input, and the residual certificate for the boundary value, sharpness 1.0000–2.4937, 16/16 in budget), the three T110 pieces of the propagation circle (the exact splitting of the growth step, embedding error $\leq 2.6 \times 10^{-14}$, with the new atom's restriction to the old window the exact zero matrix; the graded Loewner minorant step, valid exactly under the induction hypothesis, holding 15/15 with retention 1.0000; and the base-case Cholesky certificate at $n = 2$, $\lambda_{\min} \geq 6.93 \times 10^{-2}$ at three resolutions), and their T111 deep-ladder extension (117 handover certificates at retention 1.000000, largest single loss 5.96×10^{-8}; atom entry the exact zero matrix on 117/117; and the arithmetic ladder-wall statement at frozen cell width, gap $> 2D$ needed against the twin-prime gap 0.003831 at $521 \to 523$), the T114 margin-free step certificates (Albert/pseudo-inverse Schur, 27/27 ladder steps to $n = 1331$ including all seven torn zones, with the exact 4×4 Schur complement at 42–67% of the block scale — no margin division anywhere), and the T115 compression certificates (the exact identity $X_{\text{mixed}} = Q_{\text{old,mixed}}$; Albert on 66/66 (zone, q) combinations; the one-sided second-order compression error; the deep margin-free step at $n = 155\,921$ and the ten-step chains — certificates for a coarser Galerkin space, declared as such), and the T116 boundary-recursion identities (the partial-minimisation identity $t^\top T^{-1} t = r^\top \Sigma^{-1} r + \sigma$ on six zones; the Haynsworth double-Albert equivalence 6/6 with sharp negative controls; the exact Riccati chaining, bit-exact on real windows; and the exact carriage of the global rank-one pole in the bordered state — the deep march itself certifies the band model's cost geometry, declared as such), and the T117 epsilon-theorem package (the four Galerkin identities — the cell functional of $2\sinh(x/2)$, the exact nesting $T_c = P^\top T_f P$, the two-level Pythagoras, the dual-norm residual form, all to 10^{-10} or better; the two-level lower-bound chain, every link an identity or a completed Cholesky, positive on all 19 nested pairs; and the exact jump formulas — the Levinson bordering drop r_0^2/s_0 and the rank-(k_0+1) Woodbury corner update, both to 10^{-10}), and the T118 symbol package (the closed-form two-grid symbols — σ_S the weighted harmonic and σ_z the arithmetic mean of the aliasing pair; the saturation identity $\varepsilon_c - \varepsilon_f = y^\top S y$ to 4.9×10^{-11}; the CBS identity $\lambda_{\min}(L_z^{-1} S L_z^{-\top}) = 1 - \gamma^2$ to 2.1×10^{-15}; the isometry ordering onto $T_h(\sigma_z)$; $\inf \sigma_z > 0$ with explicit second-order margins on 7/14 windows; and the pole-mass identity in the oscillation space, $\leq 2.1 \times 10^{-5}$ via the DTFT factor $|\sin(\theta/2)|$).

Measured (holds on the resolved windows; no uniform constant): ρ_-; the ratio $r = 0.0053$–0.1814 and $\hat{r} < 1$ at all 92 resolution points; $\omega = 0.2366$–0.7531; the strict odd-channel margin $\lambda_{\min}(Q|_{\text{odd}}) = 1.4 \times 10^{-5}$–$5.1 \times 10^{-2}$; the margin map $m_k \geq$ need109 on 16/16 zones (ratio 2.09–179.6), extended by T111 to 199 zones with the crossing *measured* between $n = 461$ and $n = 463$ ($n^* \approx 462$, an upper bound; the T110 extrapolation $n \approx 170$ was a fit artefact); the reserve trend $f_{\text{crit}} \sim n^{-0.39}$ (fit); the requirement wall $n = 727$; the closure map 44/64; the handover margins; the two race slopes; the T114 ratio closure $r = 2 \times 10^{-4}$–0.103 on 27/27 steps and the D-powers of ε and κ (fits); the T115 transport split (the bracket's lower end positive on 7/11 regrid pairs, threshold $\rho^* = 1.83$ against a median ratio 2.001) and the fall laws $\lambda_{\min}(S) \sim \rho^{-1.69}$, $\varepsilon \sim \rho^{-1.71}$ (fits); the T116 law $\varepsilon \approx 8.34\, D^{1.790}\alpha^{-6.04}$ (jackknife ≤ 0.041), the factor-120 jumps at prime-power entries, the comb truncation budget growing like $n^{2.17}$, and the criticality $\tilde{t}^{\mathsf{T}}T^{-1}\tilde{t} \to 0.999975745$ (fits/measurements); the T117 rates $\theta'(\text{sum}) = 1.74$ against $\theta = 1.76$ (loss 0.04 powers of D) with bound/$\varepsilon \in [0.111, 0.185]$, the growth of $\lambda_{\min}(S)$ under refinement, the entry direction (all 23 prime-power entries *raise* ε — correcting the T116 factor-120 reading, a sweep artefact), and the saturation ratio in $[0.675, 0.744]$; the T118 measurements: the corner-exponent drift $+0.238$ per halving (3–7.4σ), the concentration growth $\sim h^{0.99}$, the CBS floor $1 - \gamma^2 \geq 0.181$, the computed saturation band $\beta \in [0.252, 0.336]$ on 20 triples, the log-versus-power growth decision on the 128× FFT lever (fits with jackknife bands), the reduction sharpness 32–76% on 14/14 windows, and the zero crossings of $\inf \sigma_z$ on 3/5 zones; every scaling exponent quoted in this document.

Hypothesis input / classical (used, named, not proved here): Kneser, Witt/Eichler, Shimura, Waldspurger/Baruch–Mao, Cohen, Weil–Guinand, Bombieri, Yoshida, Connes–Consani, Douglas, Slepian–Landau–Pollak, Gronwall/Robin (unconditional form only), Grenander–Szegő, Sherman–Morrison/Woodbury, Christoffel–Darboux, Szegő–Levinson, the Weyl bound, Combes–Thomas/Jaffard, Prager–Synge, first-order Kato perturbation, Albert (the pseudo-inverse Schur criterion), Haynsworth (the partial-minimisation form of the Schur complement), Céa/Strang's first lemma, Bramble–Hilbert, Brandt/Yserentant two-scale spaces, Szegő–Kolmogorov, Aubin–Nitsche duality, Payne–Weinberger, Levinson/Durbin, Kac–Murdock–Szegő/Widom/Fisher–Hartwig, Fejér–Riesz, the local Fourier / two-grid symbol (Brandt, Hackbusch, Fiorentino–Serra), Yserentant's strengthened Cauchy–Schwarz constant, the Bank–Smith/Dörfler–Nochetto saturation assumption (with Bornemann–Erdmann–Kornhuber as the honest caveat that saturation can fail), Wilkinson/Higham backward error.

9.2 The asymptotic mountain

Three separate reasons why the cascade of Figure 8 is not a route to a theorem as it stands, and why the document does not present it as one. Since T110 the mountain has a *precise* formulation — the three gaps of Section 7: no reserve, no scalar step law (structurally excluded by the boundary layer), and no uniformity in k. Since T111 it also has a *measured* one — the three walls: the margin wall at $n^* \approx 462$ (measured, an upper bound; the earlier $n \approx 170$ was a fit artefact), the arithmetic ladder wall at the twin-prime pair $521 \to 523$, and the requirement wall at $n = 727$. The mechanism itself never failed on the deep ladder — all 117 handovers certify at retention 1.000000, and the reserve even opens with depth ($f_{\text{crit}} \sim n^{-0.39}$, fit) — so the actionable content of the measurement is the reinterpretation: the operating variable is the window depth, not the zone index, and a proof along this route must beat the exponent gap $-0.974 = -0.681 - 0.293$ and couple the depth δ_0 to the local prime gap. T112 (`adaptive_scaling_probe.py`) then performed that coupling: the ladder wall and the depth dependence of the requirement wall fall *structurally* in the gap-coupled frame, but the margin wall is frame-invariant at exponent -0.974, with the prime-gap dependence disclosed (Bertrand–Chebyshev suffices upward; Zhang/Maynard force θ-uniformity downward). T113 (`limit_operator_probe.py`) then settled what that invariant number *is*: the wall carries the same exponent in every admissible currency — it is substance — but it measures the *discretisation*, not the spectrum: at a fixed window both leading eigenvalues carry the same cell-width power under refinement (the continuum window form has no gap), the positive floor is a cancellation of relative size 10^{-7}–10^{-4}, and the T109 requirement chain divides by an artifact margin. T114 (`margin_free_step_probe.py`) then discharged the leading part of that balance: the margin-free step certificate exists (Albert 1969, the exact Schur complement), runs eleven steps past the old wall to $n = 1331$, certifies all seven torn zones, and shows the wall was an $O(1)$ numerator divided by an artifact floor — chains now stop only at the $h \leq 1500$ window cost, never at a step, and the (R) demand itself is grid-bound and deleted ([P5]). T115 (`schur_transport_probe.py`) then split the two-object core: the transport of the exact Schur complement across a real regrid certifies only mild refinement ($\rho \lesssim 1.8$) — and for a principled reason, since on nested ladders, where the transport error is exactly zero, $\lambda_{\min}(S)$ itself falls like $\rho^{-1.7}$ — while the two-scale compression breaks the compute cap: a certified margin-free step at $n = 155\,921$ (117× deeper), ten-step chains, the stopper always cost and never a step, with the contiguous reach moving from $n \le 125$ to $n \le 5437$ (the $n \log n$ barrier divided, not broken). T116 (`boundary_formulation_probe.py`) then ran the boundary reformulation itself: the induction step *is* a boundary process — the global pole rides exactly in a $12 \times 12 + 12 + 1$ state and one unbroken Riccati march ran 169 236 prepends to 1.35 million cells (903× the old cap) at flat cost, declared as a cost-geometry demonstration and not a Weil certificate — but the state cannot be bounded, because the prime comb itself refuses compression (every incoming cell reflects off every prime power in the window; the truncation budget grows like $n^{2.17}$). T117 (`epsilon_theorem_probe.py`) then made that inequality theorem-shaped: the Galerkin reading is four exact identities, the lower-bound direction is a certified two-level chain that loses no power of D ($\theta' = 1.74$ against $\theta = 1.76$), the jumps have exact closed forms — and, correcting T116, the prime-power entries *raise* ε, so the factor-120 falls were a sweep artefact and the jump side-condition disappears. Since T117 the mountain is three named analytic lemmas about one symbol, each with a classical address — the Kac–Murdock–Szegő/Widom corner asymptotics of T^{-1}, a one-level lower bound for $\lambda_{\min}(S)$ at the Nyquist frequency, and the Bank–Smith saturation constant, two of the three constants rather than rates — plus the comb as the honest support obstruction. T118 (`symbol_lemmas_probe.py`) then measured the three against their addresses, and two of the three stand: saturation is an *identity* here; the Schur floor is rescued by the strengthened Cauchy–Schwarz shift onto the oscillation Gram, certified on 7/14 windows with the remainder shown to be under-resolved rather than obstructed on the 15 680-point FFT lever; and the corner lemma corrects itself — a log singularity is the boundary of all powers, so the chain is repaired at its last step and the rate now lives on the oscillation mass $\|y\|^2$. Since T118 the mountain is *one* genuinely new analytic statement — (H2), a $D^{1.75}$ lower bound for the oscillation mass — plus the standard CBS uniformity (H1), two arithmetic reductions for the comb dips, and the **[SUPPORT]** obstruction. T119 (`oscillation_mass_probe.py`) was running at the time of writing.

1. **Sixteen zones are sixteen zones.** Everything below the top line of the cascade is established on the resolved zones $n = 2 \ldots 29$ (T111's deep ladder extends the measurements to $n = 521$, T114's margin-free ladder to $n = 1331$, T115's compressed run to $n = 155\,921$, T116's boundary march to 1.35×10^6 cells with atom reach $n = 173$ — finitely many zones either way). Not one constant in the chain is certified uniformly in k. The T101 requirement list (A)–(D) is unchanged: the hardness would sit in (A), a sharp, non-perturbative arithmetic lower bound at a Weil-positivity edge.

2. **Finite resolution is a wall, and a measured one.** Numerics as a witness is exhausted beyond $\alpha \approx 0.55$ ($O(M^{-1.8})$, T96); the finite-block route was judged the structurally wrong instrument, not an under-resourced one (T92); $\lambda_{\min}$ collapses geometrically by a factor 11.4 per mode, so double precision ends at $N \approx 10$ while $N = 24$ would need 10^{-27} entries.

3. **The induction base is not free either.** The recursion terminates arithmetically (240/240 in ≤ 4 steps to the classical zone, T99; 960/960 with longest chain 13 steps, T101), but the zone peaks need percent-level bounds, not 0.1% (T99: extrapolated peak margins $+2.7 \times 10^{-2}$, $+1.4 \times 10^{-2}$, $+1.7 \times 10^{-2}$, i.e. 2.2–5.5% of $\mu_k/2$, with zone 5 saturating exactly).

9.3 Extraction candidates: the community-checkable path

The useful question at this point is not "how close is the program to RH" — it is not close, and no such claim is made — but "which pieces of it are unconditional theorems that someone else can check". Two candidates are named in the diary and are repeated here as the outlook:

- **Zone extension beyond** $\log 2$**.** Unconditional Weil positivity is classically known exactly up to support width $\log 2$ (Yoshida 1992; Bombieri 2000). T91's target inequality — for $a \in (\log 2, a^*]$ and $\|f\| = 1$, $(P_{\text{pole}} + A_{\text{arch}})(f) \ge \sqrt{2} \log 2 \cdot h_f(\log 2)$ — is typed as *provable-shaped* and would, if proved, be a standalone classical result: a zone extension past Bombieri's boundary. Its typing is explicit and unflattering: RH $\Rightarrow$ (T), but (T) $\not\Rightarrow$ RH. T92 showed the finite-block route to it is the wrong instrument; that narrows, rather than closes, the target.

- **The matching lemma.** Already proved exact-integer on $[4, 10^6]$ with an explicit margin inside v541, and reduced by T77 to a restricted σ_{-1} divisor bound in the Gronwall regime — with the note that Robin's *unconditional* 1983 bound suffices at the measured headroom, so the RH-equivalent sharpening is not needed. Closing the tail (the correlated-cancellation lemma on credit-rich non-coherent atoms) would be a self-contained classical statement.

Final fence

The prime line of TFPT has produced seven machine-checked modules about lattice-native Hecke structure, a finite relative-trace identity, and an exactly closing transport ledger. It has *not* produced evidence for the Riemann Hypothesis, and the one place where RH appears — I5 — appears as an equivalence typing that names the residual difficulty rather than reducing it. The corresponding open gates in the ledger (`ZETA.CENSUS.TO.GL1`, `ZETA.HP.CARRIER`) are untouched by everything in Sections 5–7, and no marker moves anywhere on account of this document.

A Probe index, diary parts T11–T118

Every row is one exploratory probe in `experiments/tfpt-discovery/`; the file column omits the common suffix `_probe.py`. "Checks" is the probe's own assertion count, all green. Verdict strings are the diary's preregistered verdict enums. Rows marked "→ vN" were later consolidated into the corresponding load-bearing module; the promotion, not the probe, is the load-bearing artefact. The index covers the 108 parts T11–T118 that the current diary file records (109 runs — T104 was a double run with two independent arms — and 2776 green checks); the ten earlier probes T1–T10 predate that file and are not indexed here. Series totals as recorded by the diary at T118: 118 **probes,** 2968/2968 **sandbox checks**, plus v535–v541 promoted. T119 (`oscillation_mass_probe.py`, contract `OSCILLATION.MASS`) was still running when this document was written and is deliberately absent.

Part	Probe	Contract / finding	Verdict	Checks
T11	`e8_glue_chi4_signed_theta`	the χ_4-signed glue theta welds parts 6 and 7	NO PROMOTION	19/19
T12	`e8_glue_lseries_selfdual`	L-series of the signed glue count; Apéry/$\zeta(3)$ bridge; Poisson self-dual width 2π	NO PROMOTION	16/16
T13	`e8_glue_character_atlas`	glue-character atlas of E_8	NO PROMOTION	11/11
T14	`seam_selfdual_width`	SEAM.SELFDUAL.WIDTH; unimodularity gate	DEFLATED	13/13
T15	`zeta_arith_completion`	ZETA.ARITH.COMPLETION; shell Hamiltonian	PARTIAL	12/12
T16	`zeta_local_euler`	ZETA.LOCAL.EULER; one weight-4 Hecke polynomial	PARTIAL	16/16
T17	`zeta_funceq_duality`	ZETA.FUNCEQ.DUALITY; Poisson involution	PARTIAL	9/9
T18	`zeta_weil_recovery_trace`	ZETA.TRACE.COMPLETENESS; Guinand–Weil balance	BENCHMARK-STANDING	17/17
T19	`prime_thinning_seam_truncation`	PRIMON.SEAM.TRUNCATION; thinning vs. the finite tower	MIXED	19/19
T20	`seam_archimedean_kernel`	archimedean kernel from the seam density of states	KILLED-AS-NAIVE	25/25
T21	`prime_prediction_mechanics`	four prediction channels quantified	MAPPED	19/19
T22	`seam_halfcut_dilation_arch`	non-DOS successor of the T20 kill	PARTIAL (K1, K2 pass; K3 fails)	25/25
T23	`seam_boost_arch_dictionary`	arch dictionary via the seam boost	DEAD	23/23
T24	`seam_truncated_trace_rvm`	truncated trace / Riemann–von Mangoldt route	DEAD	25/25
T25	`seam_tate_phase`	scattering phase as the last observable	DEAD (arch route closed)	29/29

Part	Probe	Contract / finding	Verdict	Checks
T26	`shell_multiplicity_one_quotient`	shell quotient towards multiplicity one	PARTIAL	18/18
T27	`e8_pneighbor_hecke`	Kneser p-neighbours carry the Eisenstein half	PARTIAL → v535	27/27
T28	`e8_kneser_transport_holonomy`	cusp coefficient from mod-3 lattice geometry	TRANSPORT-VISIBLE	22/22
T29	`primon_transport_category`	primon monoid relations	RELATIONS-VERIFIED	16/16
T30	`e8_transport_cusp_uniformity`	uniformity of the T28 reading at $p = 5, 7$	BROKEN (reading); transport holds	32/32
T31	`e8_neighbor_operator_decomposition`	cusp extraction is a rule $\nu_p = a\,\mathrm{Id} + b\,T_p$	REPAIRED-BY-DICTIONARY → v535	49/49
T32	`census_newform_recovery`	census redundancy is 2-adic oldform structure	REDUNDANCY-IS-OLDFORM → v535	24/24
T33	`e8_nu_rule_many_primes`	hardening of the ν-rule to $p \le 100$; Eichler finding	HARDENED → v536	26/26
T34	`compiler_functor_theta`	project ZETA.COMPILER.FUNCTOR founded	FOUNDED-WITH-ANCHOR	13/13
T35	`rankin_selberg_functor`	Rankin–Selberg translates the abelian shadow	RS-TRANSLATES-EISENSTEIN	13/13
T36	`e8_lambda_charsum_evaluator`	cusp remainder isolated geometrically	TWO-SIDED → v536	29/29
T37	`e8_salie_signed_cusp`	signed cusp channel, scalable	SCALABLE-AND-SIGNED	37/37
T38	`cuspidal_bridge`	Shimura preimage of f_8 inside the compiler monoid	BRIDGE-FOUND → v537	13/13
T39	`zeta_center_atlas`	the abelian drop	ABELIAN-DROP-CLOSED	13/13
T40	`weil_recovery_compiler_sector`	Weil-recovery sector terrain	TERRAIN-MAPPED	24/24
T41	`functor_wiring_halfintegral`	the half-integral bridge is Hecke-equivariant	FUNCTOR-WIRED → v537	13/13
T42	`e8_local_densities`	local densities closed	CLOSED → v536	25/25
T43	`kohnen_plus_waldspurger`	Kohnen plus space	PARTIAL	12/12
T44	`kohnen_operator_projection`	central-value route via Kohnen plus	OUTSIDE-KOHNEN-SCOPE → v537	17/17
T45	`baruch_mao_metaplectic`	central-value family as $b(d)^2$; R constant	CENTRAL-VALUES-WIRED → v537	13/13
T46	`stage4_prolate_prototype`	stage-4 prolate prototype	IMPL-OPEN	16/16
T47	`stage4_prolate_uv_dirac`	pointwise UV/Dirac prototype	UV-NO-SIGNAL	17/17
T48	`lambda_scale_uniqueness`	does the compiler fix the Λ scale?	SCALE-TORSOR (claim killed)	17/17
T49	`rankin_positivity_miniature`	the Deligne/Rankin mechanism runs in-suite	MINIATURE-RUNS	28/28
T50	`half_integral_selfconvolution`	the central-value family aggregates exactly	FAMILY-AGGREGATED	17/17
T51	`waldspurger_family_kernel`	first non-collapsing infinite carrier candidate	INFINITE-CARRIER-CANDIDATE	22/22
T52	`hecke_orbit_transfer`	path space of Hecke words; fibre determinants	TRANSFER-REALIZED	16/16
T53	`relative_trace_compiler`	v535/v536/v537 are three projections of one RTF identity	ONE-FORMULA → v538	16/16
T54	`waldspurger_canonical_measure`	the “canonical discriminant measure” claim	MARGINAL-WEIGHT (killed)	26/26
T55	`rtf_period_pairing`	canonical distributional limit of the finite RTF	DISTRIBUTIONAL-RTF-TYPED → v539	36/36
T56	`rtf_pairing_sharpening`	all T55 identifications to sub-percent	SHARPENED	27/27
T57	`infinite_rtf_packing`	both infinity directions in one double series $Z(s, w)$	PACKED	24/24

Part	Probe	Contract / finding	Verdict	Checks
T58	`zeta_rtf_xi_residue`	GL(1) shadows of $Z(s,w)$	GL1-SHADOW-CLASSICAL (route A closed)	24/24
T59	`weil_gns_identification`	residual factor in closed form	TRANSFORM-REQUIRED	19/19
T60	`plancherel_descent`	Sato–Tate moment contraction	PARTIAL	21/21
T61	`st_isotype_gl1_core`	the trivial Sato–Tate isotype is a GL(1) core	GL1-CORE-EXACT → v539	24/24
T62	`core_isolation`	d-twist mixing vanishes fibrewise	CORE-ISOLATED → v539	24/24
T63	`weil_core_completion`	Plancherel correction irreducibly non-automorphic	EXTRA-TERMS → v539	33/33
T64	`rtf_stabilization`	$\det_2$ absorbs obstruction 2; the minus stays	HALF-STABLE	34/34
T65	`pi_digits_prime_distribution`	prime distribution in the digits of π	PI-NULL	16/16
T66	`pi_prime_four_level`	the π/prime question on four levels	FOUR-LEVEL-NULL	23/23
T67	`amplitude_dirac_sqrt`	Dirac square root of the amplitude plane	DIRAC-SQRT-EXACT → v540	27/27
T68	`geometric_polarization`	geometric polarisation $b = N_+ - N_-$	RESHUFFLE-ONLY → v540	34/34
T69	`mixed_channel_full_weight`	even-k deletion is a determinant invariant	DELETION-UNIVERSAL → v540	33/33
T70	`linear_measure_weil`	positive linear Θ carrier; pure plus combination	LINEAR-PLUS-COMBINATION → v540	29/29
T71	`transport_terrain`	FE of the linear family exactly anchored	TERRAIN-MAPPED → v540	40/40
T72	`cone_enlargement`	the cone library saturates at 5/24	COMPLEMENTARY-PAIR → v540	34/34
T73	`continuous_cone`	per-direction hybrid cones in the continuum	HYBRID-GAINS	28/28
T74	`door_a_spectral_polarization`	the spectral side is provably sign-blind	VACUUM-STRUCTURE	32/32
T75	`door_b_lambda_transport`	the gap functional λ^* has its own structure	LAMBDA-STRUCTURED	28/28
T76	`hybrid_universality`	the recipe certifies 91/91 adversarial Weil directions	RECIPE-UNIVERSAL-ON-BATTERY	22/22
T77	`matching_lemma_structure`	the matching lemma reduces to a restricted divisor bound	LEMMA-CLASSICAL-SHAPED	21/21
T78	`lemma_window_proof`	matching lemma exact-integer on $[4, 10^6]$	WINDOW-PROVED → v541	25/25
T79	`transport_ledger`	the transport ledger closes; I5 named and typed	LEDGER-CLOSES → v541	22/22
T80	`tail_correlation_lemma`	character-exact signed envelope; tail to $N^* \approx 10^{23}$	RESERVE-PARTIAL → v541	29/29
T81	`recipe_coherent_avoidance`	reachability theorem: coherent targets need coherent m	AVOIDANCE-FAILS → v541	31/31
T82	`heat_arch`	Δ_{arch} is the internal Γ difference	ARCH-INTERNAL → v541	22/22
T83	`fe_symmetrization`	symmetrisation absorbed; the wall is FE-transversal	INVARIANT-NULL	27/27
T84	`gaussian_lift`	Grössencharacter phases give $L(1, \lambda_k)$ convergence	LIFT-WORKS-UNANCHORED	29/29
T85	`lambda_equivariant_design`	the λ design closes the coherent class	LEMMA-CLOSES-LAMBDA → v541	24/24
T86	`correlated_cancellation`	the Q-pairing involution closes the non-coherent tail	LEMMA-FULLY-CLOSED	29/29
T87	`i5_connes_dictionary`	I5 one-family form vs. the semi-local Connes structure	DICTIONARY-EXACT-CORE	22/22
T88	`i5_tightness`	eight near-vertical tight curves; Γ-side spacing	TIGHT-SET-PARAMETRIZED	27/27
T89	`sonin_crossing`	window compression is Bombieri's classical object	CROSSING-MAPPED	19/19
T90	`residual_dissection`	the residual vectors are explicit Gaussian modes	CORE-DISSECTED	17/17
T91	`thin_band_analytic`	the band is the one-atom zone; atom rescue	BAND-PARTIAL (3/4 closed)	19/19
T92	`zone_extension_proof`	the finite-block route is the wrong instrument	T-SKELETON	36/36

Part	Probe	Contract / finding	Verdict	Checks
T93	`band_reconciliation`	third independent implementation of the T89/T91 band map	MIXED (map recalibrated)	41/41
T94	`prime_blind_demo`	753/753 primes of an unseen window from lattice counting	BLIND-100	16/16
T95	`continuum_extension`	C1 chain proved; band top behaves as a positivity edge	T-CONTINUUM-NUMERIC	28/28
T96	`relay_test`	the T95 edge was a map artefact; the relay is real	EDGE-ARTIFACT + RELAY-CONFIRMED	21/21
T97	`relay_induction`	the induction step has proof shape; one cross-block lemma	ALIGNMENT-ONLY	105/105
T98	`cross_lemma`	the $\sqrt{\lambda_0}$ law is Douglas range inclusion (circular)	LAW-CONFIRMED-MECHANISM-OPEN	44/44
T99	`dk_recursion`	exact mirror-parity selection rule; recursive inequality	DECAY-LAW-FOUND	23/23
T100	`bulk_remainder`	closure 11/24 → 24/24 samples; the drift was a grid artefact	REMAINDER-CLOSES-ZONES	27/27
T101	`k_asymptotics`	16 zones resolved; primitives flat, instrument loses the race	CROWDING-TRENDS	31/31
T102	`arithmetic_bound`	the onset is manufactured by the Schur dressing; C/g was a proxy	MECHANISM-IDENTIFIED	42/42
T103	`instrument`	slope halves; closure map 7/64 → 44/64	INSTRUMENT-IMPROVED	29/29
T104	`schur_profile_bound`	arm B: naive margin route dead, threshold split closes 16/16	CHAIN-PARTIAL	21/21
T104	`schur_profile_chain`	arm A: independent twin of the same contract	CHAIN-PARTIAL	47/47
T105	`bare_wing_bound`	bare bound certified in closed form; avoidance is a theorem	ONE-OF-TWO	28/28
T106	`friedrichs_angle`	parity split of the Weil pole; hardness in the odd channel	DENSITY-MAPPED	32/32
T107	`odd_channel_closure`	Sherman–Morrison/Woodbury reduction to $r = \kappa/\varepsilon$	SCALAR-TRACTABLE	30/30
T108	`ratio_certificate`	ε is an exact $Q\|_{\mathrm{odd}}$ energy; (R) on two scalars; first non-vacuous margin chain	EPSILON-IDENTITY	44/44
T109	`boundary_decay`	cancellation, not decay; both scalars certified (graded cap + residual certificate); one strict-margin input left	BOUNDARY-CERTIFIED	29/29
T110	`margin_propagation`	the induction circle closes on the measured zones (certified base case + graded Loewner minorant; atom entry free); three gaps named	MARGIN-PROPAGATES	28/28
T111	`deep_zone_stress`	the crossing measured, not extrapolated ($n^* \approx 462$, upper bound); three walls (margin / twin-prime ladder / requirement); mechanism never fails, 117/117	CROSSING-CONFIRMED	23/23
T112	`adaptive_scaling`	gap-coupled frame: ladder wall and ω depth-dependence fall structurally; margin wall frame-invariant (-0.974); trade to [P1]/[P2], prime-gap inputs disclosed	SCALING-PARTIAL	20/20
T113	`limit_operator`	the wall is currency-invariant (-1.168 in all five currencies) but measures the discretisation: no continuum gap, floor a 10^{-7}–10^{-4} cancellation, need109 an artifact margin; new balance [P1]–[P4]	SUBSTANCE-CONFIRMED	27/27
T114	`margin_free_step`	the margin-free step: Albert/pseudo-inverse Schur certifies 27/27 steps (11 past the old wall, to $n = 1331$) and all 7 torn zones; the wall was an $O(1)$ numerator over an artifact floor; (R) grid-bound, deleted ([P5]); new two-object core	WALL-DISSOLVES	22/22
T115	`schur_transport`	transport certifies only mild refinement (split at $\rho^* \approx 1.8$; on nested ladders $\lambda_{\min}(S)$ itself falls like $\rho^{-1.7}$), but the two-scale compression breaks the cap: margin-free step at $n = 155\,921$, ten-step chains, stopper always cost; remaining list three points, one inequality (Szegő–Levinson)	TRANSPORT-BLOCKED	26/26

Part	Probe	Contract / finding	Verdict	Checks
T116	`boundary_formulation`	the induction as a boundary process: the (Σ, r, σ) state carries the global pole exactly and marches 169 236 prepends to 1.35×10^6 cells at flat cost (cost-geometry demo, not a Weil certificate); the prime comb refuses compression (budget $\sim n^{2.17}$); $\varepsilon \approx 8.34\, D^{1.790}\alpha^{-6.04}$ with factor-120 jumps — a Galerkin error under Aubin–Nitsche duality	RICCATI-PARTIAL	33/33
T117	`epsilon_theorem`	the one inequality theorem-shaped: four exact Galerkin identities ($\varepsilon > 0 \iff \tilde{t} \notin T(V_c)$); a certified two-level lower bound on 19/19 pairs at $\theta' = 1.74$ against $\theta = 1.76$ (no power of D lost); exact jump formulas (Levinson bordering, rank-(k_0+1) Woodbury) — entries *raise* ε, the factor-120 falls were a sweep artefact; three named analytic lemmas remain, two of them constants	THEOREM-SHAPED	23/23
T118	`symbol_lemmas`	the three lemmas against their classical addresses: saturation is an *identity* here ($\varepsilon_c - \varepsilon_f = y^\mathsf{T} S y$, $\beta \in [0.252, 0.336]$); the S-symbol is a weighted *harmonic* aliasing mean, vacuous on the comb — rescued by the CBS shift onto the oscillation Gram, whose *arithmetic* mean suppresses the comb quadratically ($\inf \sigma_z > 0$ on 7/14 windows; on the 15 680-point FFT lever the floor rises logarithmically and crosses zero on 3/5 zones — under-resolved, not obstructed); the corner exponent drifts (log is the boundary of all powers) and the chain is repaired on $\|y\|^2$; one analytic statement remains — (H2), $\|y\|^2 \geq c\, D^{1.75}$	TWO-OF-THREE	36/36

B Symbols used in Sections 5–7

All of these are sandbox objects; they are listed so the figures and formulas above can be read without the diary.

Symbol	**Meaning**
Q_{Weil}, Q_{cert}, Δ_2, Δ_{arch}	the four terms of the transport ledger; $Q_{\mathrm{Weil}} = Q_{\mathrm{cert}} + \Delta_{\mathrm{arch}} + \Delta_2$ closes at 4.4×10^{-16} (v541)
A_{fam}, A_{shift}	the two archimedean halves of the heat family; $\Delta_{\mathrm{arch}} = A_{\mathrm{fam}} - A_{\mathrm{shift}}$ (T82)
I5	the one remaining coupled inequality, $Q_{\mathrm{cert}} + \Delta_2 + A_{\mathrm{fam}} - A_{\mathrm{shift}} \geq 0$ (T79)
μ_k, w_k, g_k	atom strength, handover window width, atom spacing at zone k
$D_k(\alpha)$	the Schur-complement defect of the induction step; target $D_k \leq \mu_k/2$ (T98)
$\sigma_k(\delta)$	the E_- Schur profile of the genuine Weil form just above atom entry (T102)
E_-, E_0, E_+	the three atom eigenspaces; the k-th atom acts as $\mathrm{diag}(-\frac{1}{2}, 0, +\frac{1}{2})$
bare_k, b_0	$\lambda_{\min}(Q_{\mathrm{full}}\|_{E_-})$ and its certified excess over $\mu_k/2$ (T105)
L, Λ_-	the soft dressing scalar and its free cap (5.63–7.52, divergent) (T105)
J	the mirror-parity involution; $JQJ = Q$, $JB = -BR$ (T105)
$\rho_\pm$	the Friedrichs-angle cosines of the even / odd channel (T106)
(R)	$Q_{\mathrm{full}}(\alpha_k)\|_{J=-1} \succeq (\mu_k/2)P_-\|_{J=-1}$, the remaining Loewner statement (T106)
s^*, τ, κ, ε	Sherman–Morrison scalar, its Woodbury split $s^* = \tau + \kappa$, and $\varepsilon = 1 - \tau$ (T107)
r	the residual ratio κ/ε; (R) $\iff$ $r \leq 1$; measured 0.0053–0.1814 (T107)
x, h, x_0	the pole response $x = T_{\mathrm{odd}}^{-1}\tilde{t}$, its demand compression $h = V^{\mathsf{T}}x$, and the single edge entry that carries $\\|h\\|$ on 8 zones (T108)
ω, ω_{cert}	the wing scalar $(\mu/2)\lambda_{\max}(\Gamma)$ with $\Gamma = V^{\mathsf{T}}T_{\mathrm{odd}}^{-1}V$, and its graded-matrix-cap certificate (T108/T109)
m_k, need109	the strict odd-channel margin $\lambda_{\min}(Q\|_{\mathrm{odd}})$ at zone k, and the explicit demand $(\mu/2)H_{\mathrm{cert}}^2/((1-\omega_{\mathrm{cert}})\kappa)$ of the T109 chain; $m_k \geq$ need109 on 16/16 measured zones (T110)
(Σ, r, σ); θ, φ	the boundary state of the Riccati recursion (partial minimisation of the pole-free form over the interior; carries the pole exactly), and the two exponents of the measured law $\varepsilon \approx 8.34\, D^{\theta}\alpha^{\varphi}$, $\theta = 1.790$, $\varphi = -6.04$ (T116)